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United States Patent	12414770
Kind Code	B2
Date of Patent	September 16, 2025
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Articulation assembly for a surgical device

Abstract

A surgical device includes an elongate body, a tool assembly supported on the elongate body for articulation, and an articulating mechanism for selectively articulating the tool assembly in relation to the elongate body. The articulation mechanism includes an articulation lever and a pair of locking assemblies. The articulation lever is movable to articulate the tool assembly in relation to the elongate body about an articulation axis. The locking assemblies are configured to prevent movement of the articulation lever until one of the locking assemblies is moved from a locked position to an unlocked position. The locking assemblies are movable from their locked positions to their unlocked positions prior to movement of the articulation lever to reduce the force required to manipulate the articulation lever and to facilitate articulation in continuous fashion.

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Appl. No.:	18/562392
Filed (or PCT Filed):	May 21, 2021
PCT No.:	PCT/CN2021/095023
PCT Pub. No.:	WO2022/241744
PCT Pub. Date:	November 24, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240252169 A1	Aug. 01, 2024

Publication Classification

Int. Cl.: **A61B17/072** (20060101); **F16C11/10** (20060101); A61B17/00 (20060101); A61B17/29 (20060101)

U.S. Cl.:

CPC **A61B17/07207** (20130101); **F16C11/10** (20130101); A61B2017/00367 (20130101); A61B2017/07257 (20130101); A61B2017/07271 (20130101); A61B2017/2927 (20130101)

Field of Classification Search

CPC: A61B (17/07207); A61B (2017/00367); A61B (2017/07257); A61B (2017/07271); A61B (2017/2927); F16C (11/10)

USPC: 227/176.1

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6981628	12/2005	Wales	N/A	N/A
6986451	12/2005	Mastri et al.	N/A	N/A
6988649	12/2005	Shelton, IV et al.	N/A	N/A
6991627	12/2005	Madhani et al.	N/A	N/A
6994714	12/2005	Vargas et al.	N/A	N/A
7000818	12/2005	Shelton, IV et al.	N/A	N/A
7000819	12/2005	Swayze et al.	N/A	N/A
7032799	12/2005	Viola et al.	N/A	N/A
7044352	12/2005	Shelton, IV et al.	N/A	N/A
7044353	12/2005	Mastri et al.	N/A	N/A
7055730	12/2005	Ehrenfels et al.	N/A	N/A
7055731	12/2005	Shelton, IV et al.	N/A	N/A
7059508	12/2005	Shelton, IV et al.	N/A	N/A
7070083	12/2005	Jankowski	N/A	N/A
7083075	12/2005	Swayze et al.	N/A	N/A
7097089	12/2005	Marczyk	N/A	N/A
7111769	12/2005	Wales et al.	N/A	N/A
7114642	12/2005	Whitman	N/A	N/A
7121446	12/2005	Arad et al.	N/A	N/A
7128253	12/2005	Mastri et al.	N/A	N/A
7128254	12/2005	Shelton, IV et al.	N/A	N/A
7140527	12/2005	Ehrenfels et al.	N/A	N/A
7140528	12/2005	Shelton, IV	N/A	N/A
7143923	12/2005	Shelton, IV et al.	N/A	N/A
7143924	12/2005	Scirica et al.	N/A	N/A
7143925	12/2005	Shelton, IV et al.	N/A	N/A
7143926	12/2005	Shelton, IV et al.	N/A	N/A
7147138	12/2005	Shelton, IV	N/A	N/A
7159750	12/2006	Racenet et al.	N/A	N/A
7168604	12/2006	Milliman et al.	N/A	N/A
7172104	12/2006	Scirica et al.	N/A	N/A
7188758	12/2006	Viola et al.	N/A	N/A
7207471	12/2006	Heinrich et al.	N/A	N/A
7213736	12/2006	Wales et al.	N/A	N/A
7225963	12/2006	Scirica	N/A	N/A
7225964	12/2006	Mastri et al.	N/A	N/A
7238195	12/2006	Viola	N/A	N/A
7246734	12/2006	Shelton, IV	N/A	N/A
7258262	12/2006	Mastri et al.	N/A	N/A
7267682	12/2006	Bender et al.	N/A	N/A
7278562	12/2006	Mastri et al.	N/A	N/A
7278563	12/2006	Green	N/A	N/A
7287682	12/2006	Ezzat et al.	N/A	N/A
7293685	12/2006	Ehrenfels et al.	N/A	N/A
7296722	12/2006	Ivanko	N/A	N/A

7296724	12/2006	Green et al.	N/A	N/A
7296772	12/2006	Wang	N/A	N/A
7300444	12/2006	Nielsen et al.	N/A	N/A
7303107	12/2006	Milliman et al.	N/A	N/A
7303108	12/2006	Shelton, IV	N/A	N/A
7308998	12/2006	Mastri et al.	N/A	N/A
7326232	12/2007	Viola et al.	N/A	N/A
7328828	12/2007	Ortiz et al.	N/A	N/A
7328829	12/2007	Arad et al.	N/A	N/A
7334717	12/2007	Rethy et al.	N/A	N/A
7354447	12/2007	Shelton, IV et al.	N/A	N/A
7357287	12/2007	Shelton, IV et al.	N/A	N/A
7364061	12/2007	Swayze et al.	N/A	N/A
7367485	12/2007	Shelton, IV et al.	N/A	N/A
7377928	12/2007	Zubik et al.	N/A	N/A
7380695	12/2007	Doll et al.	N/A	N/A
7380696	12/2007	Shelton, IV et al.	N/A	N/A
7396356	12/2007	Mollenauer	N/A	N/A
7398907	12/2007	Racenet et al.	N/A	N/A
7399310	12/2007	Edoga et al.	N/A	N/A
7401720	12/2007	Durrani	N/A	N/A
7401721	12/2007	Holsten et al.	N/A	N/A
7404508	12/2007	Smith et al.	N/A	N/A
7404509	12/2007	Ortiz et al.	N/A	N/A
7407074	12/2007	Ortiz et al.	N/A	N/A
7407075	12/2007	Holsten et al.	N/A	N/A
7407077	12/2007	Ortiz et al.	N/A	N/A
7407078	12/2007	Shelton, IV et al.	N/A	N/A
7416101	12/2007	Shelton, IV et al.	N/A	N/A
7419080	12/2007	Smith et al.	N/A	N/A
7419081	12/2007	Ehrenfels et al.	N/A	N/A
7419495	12/2007	Menn et al.	N/A	N/A
7422139	12/2007	Shelton, IV et al.	N/A	N/A
7424965	12/2007	Racenet et al.	N/A	N/A
7431189	12/2007	Shelton, IV et al.	N/A	N/A
7431730	12/2007	Viola	N/A	N/A
7434715	12/2007	Shelton, IV et al.	N/A	N/A
7434717	12/2007	Shelton, IV et al.	N/A	N/A
7438208	12/2007	Larson	N/A	N/A
7438209	12/2007	Hess et al.	N/A	N/A
7441684	12/2007	Shelton, IV et al.	N/A	N/A
7441685	12/2007	Boudreaux	N/A	N/A
7448525	12/2007	Shelton, IV et al.	N/A	N/A
7451904	12/2007	Shelton, IV	N/A	N/A
7455208	12/2007	Wales et al.	N/A	N/A
7455676	12/2007	Holsten et al.	N/A	N/A
7458494	12/2007	Matsutani et al.	N/A	N/A
7461767	12/2007	Viola et al.	N/A	N/A
7462185	12/2007	Knodel	N/A	N/A
7464846	12/2007	Shelton, IV et al.	N/A	N/A

7464847	12/2007	Viola et al.	N/A	N/A
7464848	12/2007	Green et al.	N/A	N/A
7464849	12/2007	Shelton, IV et al.	N/A	N/A
7467740	12/2007	Shelton, IV et al.	N/A	N/A
7472814	12/2008	Mastri et al.	N/A	N/A
7472815	12/2008	Shelton, IV et al.	N/A	N/A
7472816	12/2008	Holsten et al.	N/A	N/A
7473258	12/2008	Clauson et al.	N/A	N/A
7481347	12/2008	Roy	N/A	N/A
7481348	12/2008	Marczyk	N/A	N/A
7481349	12/2008	Holsten et al.	N/A	N/A
7481824	12/2008	Boudreaux et al.	N/A	N/A
7487899	12/2008	Shelton, IV et al.	N/A	N/A
7490749	12/2008	Schall et al.	N/A	N/A
7494039	12/2008	Racenet et al.	N/A	N/A
7500979	12/2008	Hueil et al.	N/A	N/A
7503474	12/2008	Hillstead et al.	N/A	N/A
7506790	12/2008	Shelton, IV	N/A	N/A
7506791	12/2008	Omaits et al.	N/A	N/A
7510107	12/2008	Timm et al.	N/A	N/A
7513408	12/2008	Shelton, IV et al.	N/A	N/A
7517356	12/2008	Heinrich	N/A	N/A
7537602	12/2008	Whitman	N/A	N/A
7543729	12/2008	Ivanko	N/A	N/A
7543730	12/2008	Marczyk	N/A	N/A
7543731	12/2008	Green et al.	N/A	N/A
7552854	12/2008	Wixey et al.	N/A	N/A
7556185	12/2008	Viola	N/A	N/A
7556186	12/2008	Milliman	N/A	N/A
7559450	12/2008	Wales et al.	N/A	N/A
7559452	12/2008	Wales et al.	N/A	N/A
7559453	12/2008	Heinrich et al.	N/A	N/A
7559937	12/2008	de la Torre et al.	N/A	N/A
7565993	12/2008	Milliman et al.	N/A	N/A
7568603	12/2008	Shelton, IV et al.	N/A	N/A
7568604	12/2008	Ehrenfels et al.	N/A	N/A
7571845	12/2008	Viola	N/A	N/A
7575144	12/2008	Ortiz et al.	N/A	N/A
7584880	12/2008	Racenet et al.	N/A	N/A
7588174	12/2008	Holsten et al.	N/A	N/A
7588175	12/2008	Timm et al.	N/A	N/A
7588176	12/2008	Timm et al.	N/A	N/A
7588177	12/2008	Racenet	N/A	N/A
7597229	12/2008	Boudreaux et al.	N/A	N/A
7597230	12/2008	Racenet et al.	N/A	N/A
7600663	12/2008	Green	N/A	N/A
7604150	12/2008	Boudreaux	N/A	N/A
7604151	12/2008	Hess et al.	N/A	N/A
7607557	12/2008	Shelton, IV et al.	N/A	N/A
7611038	12/2008	Racenet et al.	N/A	N/A

7617961	12/2008	Viola	N/A	N/A
7624902	12/2008	Marczyk et al.	N/A	N/A
7624903	12/2008	Green et al.	N/A	N/A
7631793	12/2008	Rethy et al.	N/A	N/A
7631794	12/2008	Rethy et al.	N/A	N/A
7635073	12/2008	Heinrich	N/A	N/A
7635074	12/2008	Olson et al.	N/A	N/A
7635373	12/2008	Ortiz	N/A	N/A
7637409	12/2008	Marczyk	N/A	N/A
7637410	12/2008	Marczyk	N/A	N/A
7641091	12/2009	Olson et al.	N/A	N/A
7641095	12/2009	Viola	N/A	N/A
7644848	12/2009	Swayze et al.	N/A	N/A
7648055	12/2009	Marczyk	N/A	N/A
7651017	12/2009	Ortiz et al.	N/A	N/A
7654431	12/2009	Hueil et al.	N/A	N/A
7658311	12/2009	Boudreaux	N/A	N/A
7658312	12/2009	Vidal et al.	N/A	N/A
7665646	12/2009	Prommersberger	N/A	N/A
7665647	12/2009	Shelton, IV et al.	N/A	N/A
7669746	12/2009	Shelton, IV	N/A	N/A
7670334	12/2009	Hueil et al.	N/A	N/A
7673780	12/2009	Shelton, IV et al.	N/A	N/A
7673781	12/2009	Swayze et al.	N/A	N/A
7673782	12/2009	Hess et al.	N/A	N/A
7673783	12/2009	Morgan et al.	N/A	N/A
7678121	12/2009	Knodel	N/A	N/A
7681772	12/2009	Green et al.	N/A	N/A
7682367	12/2009	Shah et al.	N/A	N/A
7682368	12/2009	Bombard et al.	N/A	N/A
7690547	12/2009	Racenet et al.	N/A	N/A
7694865	12/2009	Scirica	N/A	N/A
7699205	12/2009	Ivanko	N/A	N/A
7703653	12/2009	Shah et al.	N/A	N/A
7721931	12/2009	Shelton, IV et al.	N/A	N/A
7721933	12/2009	Ehrenfels et al.	N/A	N/A
7721935	12/2009	Racenet et al.	N/A	N/A
7726537	12/2009	Olson et al.	N/A	N/A
7726538	12/2009	Holsten et al.	N/A	N/A
7726539	12/2009	Holsten et al.	N/A	N/A
7731072	12/2009	Timm et al.	N/A	N/A
7735703	12/2009	Morgan et al.	N/A	N/A
7740159	12/2009	Shelton, IV et al.	N/A	N/A
7740160	12/2009	Viola	N/A	N/A
7743960	12/2009	Whitman et al.	N/A	N/A
7744628	12/2009	Viola	N/A	N/A
7753245	12/2009	Boudreaux et al.	N/A	N/A
7753248	12/2009	Viola	N/A	N/A
7757924	12/2009	Gerbi et al.	N/A	N/A
7757925	12/2009	Viola et al.	N/A	N/A

7762445	12/2009	Heinrich et al.	N/A	N/A
7766209	12/2009	Baxter et al.	N/A	N/A
7766210	12/2009	Shelton, IV et al.	N/A	N/A
7766924	12/2009	Bombard et al.	N/A	N/A
7766928	12/2009	Ezzat et al.	N/A	N/A
7770774	12/2009	Mastri et al.	N/A	N/A
7770775	12/2009	Shelton, IV et al.	N/A	N/A
7776060	12/2009	Mooradian et al.	N/A	N/A
7780055	12/2009	Scirica et al.	N/A	N/A
7784662	12/2009	Wales et al.	N/A	N/A
7789283	12/2009	Shah	N/A	N/A
7789889	12/2009	Zubik et al.	N/A	N/A
7793812	12/2009	Moore et al.	N/A	N/A
7793814	12/2009	Racenet et al.	N/A	N/A
7794475	12/2009	Hess et al.	N/A	N/A
7798385	12/2009	Boyden et al.	N/A	N/A
7798386	12/2009	Schall et al.	N/A	N/A
7799039	12/2009	Shelton, IV et al.	N/A	N/A
7810690	12/2009	Bilotti et al.	N/A	N/A
7810692	12/2009	Hall et al.	N/A	N/A
7810693	12/2009	Broehl et al.	N/A	N/A
7815090	12/2009	Marczyk	N/A	N/A
7815091	12/2009	Marczyk	N/A	N/A
7815092	12/2009	Whitman et al.	N/A	N/A
7819296	12/2009	Hueil et al.	N/A	N/A
7819297	12/2009	Doll et al.	N/A	N/A
7819298	12/2009	Hall et al.	N/A	N/A
7819299	12/2009	Shelton, IV et al.	N/A	N/A
7819896	12/2009	Racenet	N/A	N/A
7823760	12/2009	Zemlok et al.	N/A	N/A
7823761	12/2009	Boyden et al.	N/A	N/A
7824426	12/2009	Racenet et al.	N/A	N/A
7828186	12/2009	Wales	N/A	N/A
7828187	12/2009	Green et al.	N/A	N/A
7828188	12/2009	Jankowski	N/A	N/A
7828189	12/2009	Holsten et al.	N/A	N/A
7832408	12/2009	Shelton, IV et al.	N/A	N/A
7832611	12/2009	Boyden et al.	N/A	N/A
7832612	12/2009	Baxter, III et al.	N/A	N/A
7834630	12/2009	Damadian et al.	N/A	N/A
7837079	12/2009	Holsten et al.	N/A	N/A
7837081	12/2009	Holsten et al.	N/A	N/A
7841503	12/2009	Sonnenschein et al.	N/A	N/A
7845533	12/2009	Marczyk et al.	N/A	N/A
7845534	12/2009	Viola et al.	N/A	N/A
7845535	12/2009	Scircia	N/A	N/A
7845537	12/2009	Shelton, IV et al.	N/A	N/A
7845538	12/2009	Whitman	N/A	N/A
7850703	12/2009	Bombard et al.	N/A	N/A
7857183	12/2009	Shelton, IV	N/A	N/A

7857184	12/2009	Viola	N/A	N/A
7857185	12/2009	Swayze et al.	N/A	N/A
7857186	12/2009	Baxter, III et al.	N/A	N/A
7861906	12/2010	Doll et al.	N/A	N/A
7861907	12/2010	Green et al.	N/A	N/A
7866524	12/2010	Krehel	N/A	N/A
7866525	12/2010	Scirica	N/A	N/A
7866526	12/2010	Green et al.	N/A	N/A
7866527	12/2010	Hall et al.	N/A	N/A
7866528	12/2010	Olson et al.	N/A	N/A
7870989	12/2010	Viola et al.	N/A	N/A
7886952	12/2010	Scirica et al.	N/A	N/A
7891532	12/2010	Mastri et al.	N/A	N/A
7891533	12/2010	Green et al.	N/A	N/A
7891534	12/2010	Wenchell et al.	N/A	N/A
7896214	12/2010	Farascioni	N/A	N/A
7900805	12/2010	Shelton, IV et al.	N/A	N/A
7901416	12/2010	Nolan et al.	N/A	N/A
7905380	12/2010	Shelton, IV et al.	N/A	N/A
7905381	12/2010	Baxter, III et al.	N/A	N/A
7909039	12/2010	Hur	N/A	N/A
7909220	12/2010	Viola	N/A	N/A
7909221	12/2010	Viola et al.	N/A	N/A
7909224	12/2010	Prommersberger	N/A	N/A
7913891	12/2010	Doll et al.	N/A	N/A
7913893	12/2010	Mastri et al.	N/A	N/A
7914543	12/2010	Roth et al.	N/A	N/A
7918230	12/2010	Whitman et al.	N/A	N/A
7922061	12/2010	Shelton, IV et al.	N/A	N/A
7922063	12/2010	Zemlok et al.	N/A	N/A
7922064	12/2010	Boyden et al.	N/A	N/A
7926691	12/2010	Viola et al.	N/A	N/A
7926692	12/2010	Racenet et al.	N/A	N/A
7934628	12/2010	Wenchell et al.	N/A	N/A
7934630	12/2010	Shelton, IV et al.	N/A	N/A
7934631	12/2010	Balbierz et al.	N/A	N/A
7942300	12/2010	Rethy et al.	N/A	N/A
7942303	12/2010	Shah	N/A	N/A
7950560	12/2010	Zemlok et al.	N/A	N/A
7950561	12/2010	Aranyi	N/A	N/A
7950562	12/2010	Beardsley et al.	N/A	N/A
7954682	12/2010	Giordano et al.	N/A	N/A
7954683	12/2010	Knodel et al.	N/A	N/A
7954684	12/2010	Boudreaux	N/A	N/A
7954685	12/2010	Viola	N/A	N/A
7954686	12/2010	Baxter, III et al.	N/A	N/A
7954687	12/2010	Zemlok et al.	N/A	N/A
7959051	12/2010	Smith et al.	N/A	N/A
7963431	12/2010	Scirica	N/A	N/A
7963432	12/2010	Knodel et al.	N/A	N/A

7963433	12/2010	Whitman et al.	N/A	N/A
7967178	12/2010	Scirica et al.	N/A	N/A
7967179	12/2010	Olson et al.	N/A	N/A
7967180	12/2010	Scirica	N/A	N/A
7975894	12/2010	Boyden et al.	N/A	N/A
7980443	12/2010	Scheib et al.	N/A	N/A
7988026	12/2010	Knodel et al.	N/A	N/A
7988027	12/2010	Olson et al.	N/A	N/A
7988028	12/2010	Farascioni et al.	N/A	N/A
7992758	12/2010	Whitman et al.	N/A	N/A
7997468	12/2010	Farascioni	N/A	N/A
7997469	12/2010	Olson et al.	N/A	N/A
8002795	12/2010	Beetel	N/A	N/A
8006885	12/2010	Marczyk	N/A	N/A
8006887	12/2010	Marczyk	N/A	N/A
8007505	12/2010	Weller et al.	N/A	N/A
8007513	12/2010	Nalagatla et al.	N/A	N/A
8011550	12/2010	Aranyi et al.	N/A	N/A
8011551	12/2010	Marczyk et al.	N/A	N/A
8011552	12/2010	Ivanko	N/A	N/A
8011553	12/2010	Mastri et al.	N/A	N/A
8011555	12/2010	Tarinelli et al.	N/A	N/A
8012170	12/2010	Whitman et al.	N/A	N/A
8015976	12/2010	Shah	N/A	N/A
8016177	12/2010	Bettuchi et al.	N/A	N/A
8016178	12/2010	Olson et al.	N/A	N/A
8020742	12/2010	Marczyk	N/A	N/A
8020743	12/2010	Shelton, IV	N/A	N/A
8028882	12/2010	Viola	N/A	N/A
8028883	12/2010	Stopek	N/A	N/A
8028884	12/2010	Sniffin et al.	N/A	N/A
8033438	12/2010	Scirica	N/A	N/A
8033440	12/2010	Wenchell et al.	N/A	N/A
8033441	12/2010	Marczyk	N/A	N/A
8033442	12/2010	Racenet et al.	N/A	N/A
8034077	12/2010	Smith et al.	N/A	N/A
8038044	12/2010	Viola	N/A	N/A
8038045	12/2010	Bettuchi et al.	N/A	N/A
8052024	12/2010	Viola et al.	N/A	N/A
8056787	12/2010	Boudreaux et al.	N/A	N/A
8056788	12/2010	Mastri et al.	N/A	N/A
8056791	12/2010	Whitman	N/A	N/A
8061577	12/2010	Racenet et al.	N/A	N/A
8066166	12/2010	Demmy et al.	N/A	N/A
8070033	12/2010	Milliman et al.	N/A	N/A
8070034	12/2010	Knodel	N/A	N/A
8070035	12/2010	Holsten et al.	N/A	N/A
8074858	12/2010	Marczyk	N/A	N/A
8074859	12/2010	Kostrzewski	N/A	N/A
8074862	12/2010	Shah	N/A	N/A

8083118	12/2010	Milliman et al.	N/A	N/A
8083119	12/2010	Prommersberger	N/A	N/A
8083120	12/2010	Shelton, IV et al.	N/A	N/A
8087563	12/2011	Milliman et al.	N/A	N/A
8091753	12/2011	Viola	N/A	N/A
8091754	12/2011	Ehrenfels et al.	N/A	N/A
8091756	12/2011	Viola	N/A	N/A
8092493	12/2011	Marczyk	N/A	N/A
8096459	12/2011	Ortiz et al.	N/A	N/A
8096460	12/2011	Blier et al.	N/A	N/A
8100309	12/2011	Marczyk	N/A	N/A
8100310	12/2011	Zemlok	N/A	N/A
8102008	12/2011	Wells	N/A	N/A
8113405	12/2011	Milliman	N/A	N/A
8113406	12/2011	Holsten et al.	N/A	N/A
8113407	12/2011	Holsten et al.	N/A	N/A
8113408	12/2011	Wenchell et al.	N/A	N/A
8113409	12/2011	Cohen et al.	N/A	N/A
8113410	12/2011	Hall et al.	N/A	N/A
8123101	12/2011	Racenet et al.	N/A	N/A
8127975	12/2011	Olson et al.	N/A	N/A
8127976	12/2011	Scirica et al.	N/A	N/A
8132703	12/2011	Milliman et al.	N/A	N/A
8132705	12/2011	Viola et al.	N/A	N/A
8132706	12/2011	Marczyk et al.	N/A	N/A
8136713	12/2011	Hathaway et al.	N/A	N/A
8141762	12/2011	Bedi et al.	N/A	N/A
8152041	12/2011	Kostrzewski	N/A	N/A
8157148	12/2011	Scirica	N/A	N/A
8157150	12/2011	Viola et al.	N/A	N/A
8157151	12/2011	Ingmanson et al.	N/A	N/A
8157152	12/2011	Holsten et al.	N/A	N/A
8162197	12/2011	Mastri et al.	N/A	N/A
8167185	12/2011	Shelton, IV et al.	N/A	N/A
8167186	12/2011	Racenet et al.	N/A	N/A
8172121	12/2011	Krehel	N/A	N/A
8172124	12/2011	Shelton, IV et al.	N/A	N/A
8181837	12/2011	Roy	N/A	N/A
8186555	12/2011	Shelton, IV et al.	N/A	N/A
8186557	12/2011	Cohen et al.	N/A	N/A
8186558	12/2011	Sapienza	N/A	N/A
8186559	12/2011	Whitman	N/A	N/A
8186560	12/2011	Hess et al.	N/A	N/A
8193044	12/2011	Kenneth	N/A	N/A
8196795	12/2011	Moore et al.	N/A	N/A
8196796	12/2011	Shelton, IV et al.	N/A	N/A
8201721	12/2011	Zemlok et al.	N/A	N/A
8205619	12/2011	Shah et al.	N/A	N/A
8205780	12/2011	Sorrentino et al.	N/A	N/A
8205781	12/2011	Baxter, III et al.	N/A	N/A

8210412	12/2011	Marczyk	N/A	N/A
8210416	12/2011	Milliman et al.	N/A	N/A
8215532	12/2011	Marczyk	N/A	N/A
8216236	12/2011	Heinrich et al.	N/A	N/A
8220688	12/2011	Laurent et al.	N/A	N/A
8220690	12/2011	Hess et al.	N/A	N/A
8225979	12/2011	Farascioni et al.	N/A	N/A
8231040	12/2011	Zemlok et al.	N/A	N/A
8231041	12/2011	Marczyk et al.	N/A	N/A
8235272	12/2011	Nicholas et al.	N/A	N/A
8235273	12/2011	Olson et al.	N/A	N/A
8235274	12/2011	Cappola	N/A	N/A
8236010	12/2011	Ortiz et al.	N/A	N/A
8240536	12/2011	Marczyk	N/A	N/A
8240537	12/2011	Marczyk	N/A	N/A
8241322	12/2011	Whitman et al.	N/A	N/A
8245897	12/2011	Tzakis et al.	N/A	N/A
8245898	12/2011	Smith et al.	N/A	N/A
8245899	12/2011	Swensgard et al.	N/A	N/A
8245931	12/2011	Shigeta	N/A	N/A
8252009	12/2011	Weller et al.	N/A	N/A
8256653	12/2011	Farascioni	N/A	N/A
8256654	12/2011	Bettuchi et al.	N/A	N/A
8256655	12/2011	Sniffin et al.	N/A	N/A
8256656	12/2011	Milliman et al.	N/A	N/A
8267300	12/2011	Boudreaux	N/A	N/A
8272551	12/2011	Knodel et al.	N/A	N/A
8272553	12/2011	Mastri et al.	N/A	N/A
8272554	12/2011	Whitman et al.	N/A	N/A
8276594	12/2011	Shah	N/A	N/A
8276801	12/2011	Zemlok et al.	N/A	N/A
8281973	12/2011	Wenchell et al.	N/A	N/A
8286847	12/2011	Taylor	N/A	N/A
8286848	12/2011	Wenchell et al.	N/A	N/A
8286850	12/2011	Viola	N/A	N/A
8292146	12/2011	Holsten et al.	N/A	N/A
8292147	12/2011	Viola	N/A	N/A
8292148	12/2011	Viola	N/A	N/A
8292149	12/2011	Ivanko	N/A	N/A
8292150	12/2011	Bryant	N/A	N/A
8292151	12/2011	Viola	N/A	N/A
8292152	12/2011	Milliman et al.	N/A	N/A
8292153	12/2011	Jankowski	N/A	N/A
8292154	12/2011	Marczyk	N/A	N/A
8292155	12/2011	Shelton, IV et al.	N/A	N/A
8292156	12/2011	Kostrzewski	N/A	N/A
8292158	12/2011	Sapienza	N/A	N/A
8308040	12/2011	Huang et al.	N/A	N/A
8308041	12/2011	Kostrzewski	N/A	N/A
8308042	12/2011	Aranyi	N/A	N/A

8308043	12/2011	Bindra et al.	N/A	N/A
8308044	12/2011	Viola	N/A	N/A
8308046	12/2011	Prommersberger	N/A	N/A
8308757	12/2011	Hillstead et al.	N/A	N/A
8317070	12/2011	Hueil et al.	N/A	N/A
8317071	12/2011	Knodel	N/A	N/A
8322455	12/2011	Shelton, IV et al.	N/A	N/A
8322589	12/2011	Boudreaux	N/A	N/A
8328061	12/2011	Kasvikis	N/A	N/A
8328065	12/2011	Shah	N/A	N/A
8333313	12/2011	Boudreaux et al.	N/A	N/A
8336751	12/2011	Scirica	N/A	N/A
8336753	12/2011	Olson et al.	N/A	N/A
8336754	12/2011	Cappola et al.	N/A	N/A
8342377	12/2012	Milliman et al.	N/A	N/A
8342378	12/2012	Marczyk et al.	N/A	N/A
8342379	12/2012	Whitman et al.	N/A	N/A
8342380	12/2012	Viola	N/A	N/A
8348123	12/2012	Scirica et al.	N/A	N/A
8348124	12/2012	Scirica	N/A	N/A
8348125	12/2012	Viola et al.	N/A	N/A
8348126	12/2012	Olson et al.	N/A	N/A
8348127	12/2012	Marczyk	N/A	N/A
8348129	12/2012	Bedi et al.	N/A	N/A
8348130	12/2012	Shah et al.	N/A	N/A
8348131	12/2012	Omaits et al.	N/A	N/A
8353437	12/2012	Boudreaux	N/A	N/A
8353440	12/2012	Whitman et al.	N/A	N/A
8356740	12/2012	Knodel	N/A	N/A
8357174	12/2012	Roth et al.	N/A	N/A
8360294	12/2012	Scirica	N/A	N/A
8360297	12/2012	Shelton, IV et al.	N/A	N/A
8360298	12/2012	Farascioni et al.	N/A	N/A
8360299	12/2012	Zemlok et al.	N/A	N/A
8365971	12/2012	Knodel	N/A	N/A
8365972	12/2012	Aranyi et al.	N/A	N/A
8365973	12/2012	White et al.	N/A	N/A
8365976	12/2012	Hess et al.	N/A	N/A
8371491	12/2012	Huitema et al.	N/A	N/A
8371492	12/2012	Aranyi et al.	N/A	N/A
8371493	12/2012	Aranyi et al.	N/A	N/A
8381828	12/2012	Whitman et al.	N/A	N/A
8381961	12/2012	Holsten et al.	N/A	N/A
8387848	12/2012	Johnson et al.	N/A	N/A
8387849	12/2012	Buesseler et al.	N/A	N/A
8387850	12/2012	Hathaway et al.	N/A	N/A
8388652	12/2012	Viola	N/A	N/A
8393513	12/2012	Jankowski	N/A	N/A
8393514	12/2012	Shelton, IV et al.	N/A	N/A
8393516	12/2012	Kostrzewski	N/A	N/A

8397971	12/2012	Yates et al.	N/A	N/A
8397972	12/2012	Kostrzewski	N/A	N/A
8403195	12/2012	Beardsley et al.	N/A	N/A
8403196	12/2012	Beardsley et al.	N/A	N/A
8403197	12/2012	Vidal et al.	N/A	N/A
8403198	12/2012	Sorrentino et al.	N/A	N/A
8403956	12/2012	Thompson et al.	N/A	N/A
8408439	12/2012	Huang et al.	N/A	N/A
8408440	12/2012	Olson et al.	N/A	N/A
8408442	12/2012	Racenet et al.	N/A	N/A
8413868	12/2012	Cappola	N/A	N/A
8413869	12/2012	Heinrich	N/A	N/A
8413871	12/2012	Racenet et al.	N/A	N/A
8418904	12/2012	Wenchell et al.	N/A	N/A
8418905	12/2012	Milliman	N/A	N/A
8418906	12/2012	Farascioni et al.	N/A	N/A
8418907	12/2012	Johnson et al.	N/A	N/A
8418908	12/2012	Beardsley	N/A	N/A
8419768	12/2012	Marczyk	N/A	N/A
8424735	12/2012	Viola et al.	N/A	N/A
8424736	12/2012	Scirica et al.	N/A	N/A
8424737	12/2012	Scirica	N/A	N/A
8424739	12/2012	Racenet et al.	N/A	N/A
8424740	12/2012	Shelton, IV et al.	N/A	N/A
8439244	12/2012	Holcomb et al.	N/A	N/A
8439245	12/2012	Knodel et al.	N/A	N/A
8439246	12/2012	Knodel	N/A	N/A
8444036	12/2012	Shelton, IV	N/A	N/A
8444037	12/2012	Nicholas et al.	N/A	N/A
8444038	12/2012	Farascioni et al.	N/A	N/A
8448832	12/2012	Viola et al.	N/A	N/A
8453652	12/2012	Stopek	N/A	N/A
8453905	12/2012	Holcomb et al.	N/A	N/A
8453906	12/2012	Huang et al.	N/A	N/A
8453907	12/2012	Laurent et al.	N/A	N/A
8453908	12/2012	Bedi et al.	N/A	N/A
8453909	12/2012	Olson et al.	N/A	N/A
8453910	12/2012	Bettuchi et al.	N/A	N/A
8453912	12/2012	Mastri et al.	N/A	N/A
8453913	12/2012	Milliman	N/A	N/A
8453914	12/2012	Laurent et al.	N/A	N/A
8454628	12/2012	Smith et al.	N/A	N/A
8459520	12/2012	Giordano et al.	N/A	N/A
8459521	12/2012	Zemlok et al.	N/A	N/A
8459522	12/2012	Marczyk	N/A	N/A
8459523	12/2012	Whitman	N/A	N/A
8459524	12/2012	Pribanic et al.	N/A	N/A
8459525	12/2012	Yates et al.	N/A	N/A
8464922	12/2012	Marczyk	N/A	N/A
8464923	12/2012	Shelton, IV	N/A	N/A

8469252	12/2012	Holcomb et al.	N/A	N/A
8469254	12/2012	Czernik et al.	N/A	N/A
8474677	12/2012	Woodard, Jr. et al.	N/A	N/A
8479967	12/2012	Marczyk	N/A	N/A
8479968	12/2012	Hodgkinson et al.	N/A	N/A
8479969	12/2012	Shelton, IV	N/A	N/A
8485412	12/2012	Shelton, IV et al.	N/A	N/A
8490852	12/2012	Viola	N/A	N/A
8496152	12/2012	Viola	N/A	N/A
8496154	12/2012	Marczyk et al.	N/A	N/A
8496156	12/2012	Sniffin et al.	N/A	N/A
8496683	12/2012	Prommersberger et al.	N/A	N/A
8499993	12/2012	Shelton, IV et al.	N/A	N/A
8505799	12/2012	Viola et al.	N/A	N/A
8505802	12/2012	Viola et al.	N/A	N/A
8511575	12/2012	Cok	N/A	N/A
8512359	12/2012	Whitman et al.	N/A	N/A
8512402	12/2012	Marczyk et al.	N/A	N/A
8517240	12/2012	Mata et al.	N/A	N/A
8517241	12/2012	Nicholas et al.	N/A	N/A
8517243	12/2012	Giordano et al.	N/A	N/A
8517244	12/2012	Shelton, IV et al.	N/A	N/A
8523041	12/2012	Ishitsuki et al.	N/A	N/A
8523042	12/2012	Masiakos et al.	N/A	N/A
8523043	12/2012	Ullrich et al.	N/A	N/A
8534528	12/2012	Shelton, IV	N/A	N/A
8540128	12/2012	Shelton, IV et al.	N/A	N/A
8540129	12/2012	Baxter, III et al.	N/A	N/A
8540130	12/2012	Moore et al.	N/A	N/A
8540131	12/2012	Swayze	N/A	N/A
8540733	12/2012	Whitman et al.	N/A	N/A
8544711	12/2012	Ma et al.	N/A	N/A
8550325	12/2012	Cohen et al.	N/A	N/A
8556151	12/2012	Viola	N/A	N/A
8561870	12/2012	Baxter, III et al.	N/A	N/A
8561873	12/2012	Ingmanson et al.	N/A	N/A
8561874	12/2012	Scirica	N/A	N/A
8567656	12/2012	Shelton, IV et al.	N/A	N/A
8573461	12/2012	Shelton, IV et al.	N/A	N/A
8573463	12/2012	Scirica et al.	N/A	N/A
8573465	12/2012	Shelton, IV	N/A	N/A
8579176	12/2012	Smith et al.	N/A	N/A
8579177	12/2012	Beetel	N/A	N/A
8584919	12/2012	Hueil et al.	N/A	N/A
8584920	12/2012	Hodgkinson	N/A	N/A
8590762	12/2012	Hess et al.	N/A	N/A
8596515	12/2012	Okoniewski	N/A	N/A
8597311	12/2012	Criscuolo et al.	N/A	N/A
8602288	12/2012	Shelton, IV et al.	N/A	N/A
8608045	12/2012	Smith et al.	N/A	N/A

8608046	12/2012	Laurent et al.	N/A	N/A
8608047	12/2012	Holsten et al.	N/A	N/A
8613383	12/2012	Beckman et al.	N/A	N/A
8613384	12/2012	Pastorelli et al.	N/A	N/A
8616427	12/2012	Viola	N/A	N/A
8616430	12/2012	Stopek et al.	N/A	N/A
8627994	12/2013	Zemlok et al.	N/A	N/A
8628544	12/2013	Farascioni	N/A	N/A
8631988	12/2013	Viola	N/A	N/A
8631989	12/2013	Aranyi et al.	N/A	N/A
8631991	12/2013	Cropper et al.	N/A	N/A
8632525	12/2013	Kerr et al.	N/A	N/A
8632535	12/2013	Shelton, IV et al.	N/A	N/A
8636187	12/2013	Hueil et al.	N/A	N/A
8636190	12/2013	Zemlok et al.	N/A	N/A
8636192	12/2013	Farascioni et al.	N/A	N/A
8636762	12/2013	Whitman et al.	N/A	N/A
8636766	12/2013	Milliman et al.	N/A	N/A
8640940	12/2013	Ohdaira	N/A	N/A
8657174	12/2013	Yates et al.	N/A	N/A
8657177	12/2013	Scirica et al.	N/A	N/A
8657178	12/2013	Hueil et al.	N/A	N/A
8662371	12/2013	Viola	N/A	N/A
8668129	12/2013	Olson	N/A	N/A
8672206	12/2013	Aranyi et al.	N/A	N/A
8672208	12/2013	Hess et al.	N/A	N/A
8672209	12/2013	Crainich	N/A	N/A
8678263	12/2013	Viola	N/A	N/A
8678990	12/2013	Wazer et al.	N/A	N/A
8679155	12/2013	Knodel et al.	N/A	N/A
8684247	12/2013	Scirica et al.	N/A	N/A
8684249	12/2013	Racenet et al.	N/A	N/A
8684253	12/2013	Giordano et al.	N/A	N/A
8690039	12/2013	Beardsley et al.	N/A	N/A
8695865	12/2013	Smith et al.	N/A	N/A
8695866	12/2013	Leimbach et al.	N/A	N/A
8701958	12/2013	Shelton, IV et al.	N/A	N/A
8701959	12/2013	Shah	N/A	N/A
8701961	12/2013	Ivanko	N/A	N/A
8708213	12/2013	Shelton, IV et al.	N/A	N/A
8714429	12/2013	Demmy	N/A	N/A
8715277	12/2013	Weizman	N/A	N/A
8720766	12/2013	Hess et al.	N/A	N/A
8721630	12/2013	Ortiz et al.	N/A	N/A
8727197	12/2013	Hess et al.	N/A	N/A
8727200	12/2013	Roy	N/A	N/A
8733612	12/2013	Ma	N/A	N/A
8740034	12/2013	Morgan et al.	N/A	N/A
8740039	12/2013	Farascioni	N/A	N/A
8746529	12/2013	Shelton, IV et al.	N/A	N/A

8746530	12/2013	Giordano et al.	N/A	N/A
8746533	12/2013	Whitman	N/A	N/A
8746534	12/2013	Farascioni	N/A	N/A
8746535	12/2013	Shelton, IV et al.	N/A	N/A
8747421	12/2013	Balbierz et al.	N/A	N/A
8752747	12/2013	Shelton, IV et al.	N/A	N/A
8752748	12/2013	Whitman et al.	N/A	N/A
8752749	12/2013	Moore et al.	N/A	N/A
8757465	12/2013	Woodard, Jr. et al.	N/A	N/A
8758391	12/2013	Swayze et al.	N/A	N/A
8763875	12/2013	Morgan et al.	N/A	N/A
8763876	12/2013	Kostrzewski	N/A	N/A
8763877	12/2013	Schall et al.	N/A	N/A
8763879	12/2013	Shelton, IV et al.	N/A	N/A
8770458	12/2013	Scirica	N/A	N/A
8777082	12/2013	Scirica	N/A	N/A
8783541	12/2013	Shelton, IV et al.	N/A	N/A
8783542	12/2013	Riestenberg et al.	N/A	N/A
8789737	12/2013	Hodgkinson et al.	N/A	N/A
8789738	12/2013	Knodel et al.	N/A	N/A
8789739	12/2013	Swensgard	N/A	N/A
8800838	12/2013	Shelton, IV	N/A	N/A
8800840	12/2013	Jankowski	N/A	N/A
8800841	12/2013	Ellerhorst et al.	N/A	N/A
8808311	12/2013	Heinrich et al.	N/A	N/A
8814024	12/2013	Woodard, Jr. et al.	N/A	N/A
8814025	12/2013	Miller et al.	N/A	N/A
8820603	12/2013	Shelton, IV et al.	N/A	N/A
8820605	12/2013	Shelton, IV	N/A	N/A
8820606	12/2013	Hodgkinson	N/A	N/A
8820607	12/2013	Marczyk	N/A	N/A
8827133	12/2013	Shelton, IV et al.	N/A	N/A
8827134	12/2013	Viola et al.	N/A	N/A
8833631	12/2013	Munro, III et al.	N/A	N/A
8833632	12/2013	Swensgard	N/A	N/A
8840003	12/2013	Morgan et al.	N/A	N/A
8840603	12/2013	Shelton, IV et al.	N/A	N/A
8844788	12/2013	Knodel	N/A	N/A
8851354	12/2013	Swensgard et al.	N/A	N/A
8851355	12/2013	Aranyi et al.	N/A	N/A
8857693	12/2013	Schuckmann et al.	N/A	N/A
8864007	12/2013	Widenhouse et al.	N/A	N/A
8864009	12/2013	Shelton, IV et al.	N/A	N/A
8864010	12/2013	Williams	N/A	N/A
8870050	12/2013	Hodgkinson	N/A	N/A
8875971	12/2013	Hall et al.	N/A	N/A
8875972	12/2013	Weisenburgh, II et al.	N/A	N/A
8893949	12/2013	Shelton, IV et al.	N/A	N/A
8893950	12/2013	Marczyk	N/A	N/A
8899461	12/2013	Farascioni	N/A	N/A

8899462	12/2013	Kostrzewski et al.	N/A	N/A
8899463	12/2013	Schall et al.	N/A	N/A
8899464	12/2013	Hueil et al.	N/A	N/A
8900616	12/2013	Belcheva et al.	N/A	N/A
8920435	12/2013	Smith et al.	N/A	N/A
8925782	12/2014	Shelton, IV	N/A	N/A
8926598	12/2014	Mollere et al.	N/A	N/A
8931681	12/2014	Kostrzewski	N/A	N/A
8931682	12/2014	Timm et al.	N/A	N/A
8931693	12/2014	Kumar et al.	N/A	N/A
8955732	12/2014	Zemlok et al.	N/A	N/A
8958429	12/2014	Shukla et al.	N/A	N/A
8960517	12/2014	Lee	N/A	N/A
8960520	12/2014	McCuen	N/A	N/A
8967443	12/2014	McCuen	N/A	N/A
8967446	12/2014	Beardsley et al.	N/A	N/A
8973803	12/2014	Hall et al.	N/A	N/A
8978954	12/2014	Shelton, IV et al.	N/A	N/A
8978956	12/2014	Schall et al.	N/A	N/A
8979827	12/2014	Cappola	606/1	A61B 17/07207
8991677	12/2014	Moore et al.	N/A	N/A
8991678	12/2014	Wellman et al.	N/A	N/A
8998058	12/2014	Moore et al.	N/A	N/A
8998060	12/2014	Bruewer et al.	N/A	N/A
9005230	12/2014	Yates et al.	N/A	N/A
9010606	12/2014	Aranyi et al.	N/A	N/A
9010607	12/2014	Kostrzewski	N/A	N/A
9010610	12/2014	Hodgkinson	N/A	N/A
9016539	12/2014	Kostrzewski et al.	N/A	N/A
9016541	12/2014	Viola et al.	N/A	N/A
9016542	12/2014	Shelton, IV et al.	N/A	N/A
9016544	12/2014	Hodgkinson et al.	N/A	N/A
9016545	12/2014	Aranyi et al.	N/A	N/A
9016546	12/2014	Demmy et al.	N/A	N/A
9017371	12/2014	Whitman et al.	N/A	N/A
9022271	12/2014	Scirica	N/A	N/A
9027817	12/2014	Milliman et al.	N/A	N/A
9033203	12/2014	Woodard, Jr. et al.	N/A	N/A
9044228	12/2014	Woodard, Jr. et al.	N/A	N/A
9044229	12/2014	Scheib et al.	N/A	N/A
9050084	12/2014	Schmid et al.	N/A	N/A
9055941	12/2014	Schmid et al.	N/A	N/A
9060770	12/2014	Shelton, IV et al.	N/A	N/A
9072535	12/2014	Shelton, IV et al.	N/A	N/A
9084601	12/2014	Moore et al.	N/A	N/A
9084602	12/2014	Gleiman	N/A	N/A
9089326	12/2014	Krumanaker et al.	N/A	N/A
9095339	12/2014	Moore et al.	N/A	N/A
9101359	12/2014	Smith et al.	N/A	N/A

9107663	12/2014	Swensgard	N/A	N/A
9107664	12/2014	Marczyk	N/A	N/A
9107667	12/2014	Hodgkinson	N/A	N/A
9113862	12/2014	Morgan et al.	N/A	N/A
9113864	12/2014	Morgan et al.	N/A	N/A
9113870	12/2014	Viola	N/A	N/A
9113872	12/2014	Viola	N/A	N/A
9113880	12/2014	Zemlok et al.	N/A	N/A
9125649	12/2014	Bruewer et al.	N/A	N/A
9138225	12/2014	Huang et al.	N/A	N/A
9155537	12/2014	Katre et al.	N/A	N/A
9179912	12/2014	Yates et al.	N/A	N/A
9192378	12/2014	Aranyi et al.	N/A	N/A
9192379	12/2014	Aranyi et al.	N/A	N/A
9192384	12/2014	Bettuchi	N/A	N/A
9198644	12/2014	Balek et al.	N/A	N/A
9198660	12/2014	Hodgkinson	N/A	N/A
9198661	12/2014	Swensgard	N/A	N/A
9198662	12/2014	Barton et al.	N/A	N/A
9204876	12/2014	Cappola et al.	N/A	N/A
9204877	12/2014	Whitman et al.	N/A	N/A
9204880	12/2014	Baxter, III et al.	N/A	N/A
9216019	12/2014	Schmid et al.	N/A	N/A
9216020	12/2014	Zhang et al.	N/A	N/A
9220500	12/2014	Swayze et al.	N/A	N/A
9220501	12/2014	Baxter, III et al.	N/A	N/A
9220502	12/2014	Zemlok et al.	N/A	N/A
9232941	12/2015	Mandakolathur Vasudevan et al.	N/A	N/A
9232944	12/2015	Cappola et al.	N/A	N/A
9237891	12/2015	Shelton, IV	N/A	N/A
9237892	12/2015	Hodgkinson	N/A	N/A
9254180	12/2015	Huitema et al.	N/A	N/A
9265585	12/2015	Wingardner et al.	N/A	N/A
9271728	12/2015	Gupta et al.	N/A	N/A
9272406	12/2015	Aronhalt et al.	N/A	N/A
9277919	12/2015	Timmer et al.	N/A	N/A
9282962	12/2015	Schmid et al.	N/A	N/A
9283054	12/2015	Morgan et al.	N/A	N/A
9289209	12/2015	Gurumurthy et al.	N/A	N/A
9289210	12/2015	Baxter, III et al.	N/A	N/A
9289225	12/2015	Shelton, IV et al.	N/A	N/A
9295464	12/2015	Shelton, IV et al.	N/A	N/A
9295465	12/2015	Farascioni	N/A	N/A
9295466	12/2015	Hodgkinson et al.	N/A	N/A
9301752	12/2015	Mandakolathur Vasudevan et al.	N/A	N/A
9301753	12/2015	Aldridge et al.	N/A	N/A
9301757	12/2015	Williams	N/A	N/A
9301759	12/2015	Spivey et al.	N/A	N/A

9307965	12/2015	Ming et al.	N/A	N/A
9307986	12/2015	Hall et al.	N/A	N/A
9307989	12/2015	Shelton, IV et al.	N/A	N/A
9314246	12/2015	Shelton, IV et al.	N/A	N/A
9320518	12/2015	Henderson et al.	N/A	N/A
9320521	12/2015	Shelton, IV et al.	N/A	N/A
9320523	12/2015	Shelton, IV et al.	N/A	N/A
9326767	12/2015	Koch, Jr. et al.	N/A	N/A
9326771	12/2015	Baxter, III et al.	N/A	N/A
9332987	12/2015	Leimbach et al.	N/A	N/A
9345477	12/2015	Anim et al.	N/A	N/A
9345478	12/2015	Knodel	N/A	N/A
9345479	12/2015	Racenet et al.	N/A	N/A
9345481	12/2015	Hall et al.	N/A	N/A
9345780	12/2015	Manoharan et al.	N/A	N/A
9351726	12/2015	Leimbach	N/A	N/A
9351727	12/2015	Leimbach et al.	N/A	N/A
9351732	12/2015	Hodgkinson	N/A	N/A
9358003	12/2015	Hall et al.	N/A	N/A
9364217	12/2015	Kostrzewski et al.	N/A	N/A
9364218	12/2015	Scirica	N/A	N/A
9364219	12/2015	Olson et al.	N/A	N/A
9364220	12/2015	Williams	N/A	N/A
9364227	12/2015	Kostrzewski	N/A	N/A
9364231	12/2015	Wenchell	N/A	N/A
9364233	12/2015	Alexander, III et al.	N/A	N/A
9370358	12/2015	Shelton, IV et al.	N/A	N/A
9370362	12/2015	Petty et al.	N/A	N/A
9386983	12/2015	Swensgard et al.	N/A	N/A
9386984	12/2015	Aronhalt et al.	N/A	N/A
9386988	12/2015	Baxter, III et al.	N/A	N/A
9393018	12/2015	Wang et al.	N/A	N/A
9398911	12/2015	Auld	N/A	N/A
9402604	12/2015	Williams et al.	N/A	N/A
9402627	12/2015	Stevenson et al.	N/A	N/A
9421014	12/2015	Ingmanson et al.	N/A	N/A
9433419	12/2015	Gonzalez et al.	N/A	N/A
9433420	12/2015	Hodgkinson	N/A	N/A
9445810	12/2015	Cappola	N/A	N/A
9445813	12/2015	Shelton, IV et al.	N/A	N/A
9451959	12/2015	Patankar et al.	N/A	N/A
9468438	12/2015	Baber et al.	N/A	N/A
9468439	12/2015	Cappola et al.	N/A	N/A
9480476	12/2015	Aldridge et al.	N/A	N/A
9480492	12/2015	Aranyi et al.	N/A	N/A
9492171	12/2015	Patenaude	N/A	N/A
9498212	12/2015	Racenet et al.	N/A	N/A
9498219	12/2015	Moore et al.	N/A	N/A
9510827	12/2015	Kostrzewski	N/A	N/A
9517065	12/2015	Simms et al.	N/A	N/A

9517066	12/2015	Racenet et al.	N/A	N/A
9522002	12/2015	Chowaniec et al.	N/A	N/A
9539007	12/2016	Dhakad et al.	N/A	N/A
9549735	12/2016	Shelton, IV et al.	N/A	N/A
9554794	12/2016	Baber	N/A	N/A
9561029	12/2016	Scheib et al.	N/A	N/A
9561031	12/2016	Heinrich et al.	N/A	N/A
9566065	12/2016	Knodel	N/A	N/A
9572576	12/2016	Hodgkinson et al.	N/A	N/A
9579101	12/2016	Whitman et al.	N/A	N/A
9585657	12/2016	Shelton, IV et al.	N/A	N/A
9585662	12/2016	Shelton, IV et al.	N/A	N/A
9597077	12/2016	Hodgkinson	N/A	N/A
9610080	12/2016	Whitfield et al.	N/A	N/A
9615825	12/2016	Viola	N/A	N/A
9615826	12/2016	Shelton, IV et al.	N/A	N/A
9622746	12/2016	Simms	N/A	N/A
9629623	12/2016	Lytle, IV et al.	N/A	N/A
9629626	12/2016	Soltz et al.	N/A	N/A
9629628	12/2016	Aranyi	N/A	N/A
9629629	12/2016	Leimbach et al.	N/A	N/A
9642620	12/2016	Baxter, III	N/A	N/A
9649109	12/2016	Ranucci et al.	N/A	N/A
9655613	12/2016	Schaller	N/A	N/A
9662108	12/2016	Williams	N/A	N/A
9668728	12/2016	Williams et al.	N/A	N/A
9668732	12/2016	Patel et al.	N/A	N/A
9675351	12/2016	Hodgkinson et al.	N/A	N/A
9681870	12/2016	Baxter, III et al.	N/A	N/A
9687230	12/2016	Leimbach et al.	N/A	N/A
9687233	12/2016	Fernandez et al.	N/A	N/A
9693772	12/2016	Ingmanson et al.	N/A	N/A
9693815	12/2016	Motai	N/A	N/A
9700309	12/2016	Jaworek	N/A	N/A
9700319	12/2016	Motooka et al.	N/A	N/A
9706993	12/2016	Hessler et al.	N/A	N/A
9713474	12/2016	Lorenz	N/A	N/A
9717497	12/2016	Zerkle et al.	N/A	N/A
9717498	12/2016	Aranyi et al.	N/A	N/A
9724092	12/2016	Baxter, III et al.	N/A	N/A
9724095	12/2016	Gupta et al.	N/A	N/A
9757128	12/2016	Baber et al.	N/A	N/A
9763662	12/2016	Shelton, IV et al.	N/A	N/A
9770245	12/2016	Swayze et al.	N/A	N/A
9775610	12/2016	Nicholas et al.	N/A	N/A
9775635	12/2016	Takei	N/A	N/A
9782169	12/2016	Kimsey	N/A	N/A
9782173	12/2016	Mozdzierz	N/A	N/A
9795379	12/2016	Leimbach et al.	N/A	N/A
9795384	12/2016	Weaner et al.	N/A	N/A

9808246	12/2016	Shelton, IV	N/A	N/A
9808248	12/2016	Hoffman	N/A	N/A
9808249	12/2016	Shelton, IV	N/A	N/A
9820741	12/2016	Kostrzewski	N/A	N/A
9820742	12/2016	Covach et al.	N/A	N/A
9827002	12/2016	Hausen et al.	N/A	N/A
9839420	12/2016	Shelton, IV	N/A	N/A
9839421	12/2016	Zerkle et al.	N/A	N/A
9839428	12/2016	Baxter, III et al.	N/A	N/A
9855038	12/2017	Smith et al.	N/A	N/A
9855040	12/2017	Kostrzewski	N/A	N/A
9861358	12/2017	Marczyk et al.	N/A	N/A
9861359	12/2017	Shelton, IV et al.	N/A	N/A
9867612	12/2017	Parihar et al.	N/A	N/A
9867613	12/2017	Marczyk et al.	N/A	N/A
9867615	12/2017	Fanelli et al.	N/A	N/A
9872683	12/2017	Hopkins	N/A	N/A
9883860	12/2017	Leimbach et al.	N/A	N/A
9901342	12/2017	Shelton, IV et al.	N/A	N/A
9913646	12/2017	Shelton, IV	N/A	N/A
9918714	12/2017	Gibbons, Jr.	N/A	N/A
9918717	12/2017	Czernik	N/A	N/A
9924941	12/2017	Burbank	N/A	N/A
9924942	12/2017	Swayze et al.	N/A	N/A
9936951	12/2017	Hufnagel et al.	N/A	N/A
9936954	12/2017	Shelton, IV et al.	N/A	N/A
9943311	12/2017	Scirica et al.	N/A	N/A
9949737	12/2017	Zergiebel et al.	N/A	N/A
9955966	12/2017	Zergiebel	N/A	N/A
9968354	12/2017	Shelton, IV et al.	N/A	N/A
9980724	12/2017	Farascioni et al.	N/A	N/A
9987012	12/2017	Shah	N/A	N/A
9987099	12/2017	Chen et al.	N/A	N/A
9993248	12/2017	Shelton, IV et al.	N/A	N/A
9993258	12/2017	Shelton, IV et al.	N/A	N/A
10004497	12/2017	Overmyer et al.	N/A	N/A
10004505	12/2017	Moore et al.	N/A	N/A
10034668	12/2017	Ebner	N/A	N/A
10039532	12/2017	Srinivas et al.	N/A	N/A
10039545	12/2017	Sadowski et al.	N/A	N/A
10045782	12/2017	Murthy Aravalli	N/A	N/A
10052044	12/2017	Shelton, IV et al.	N/A	N/A
10058963	12/2017	Shelton, IV et al.	N/A	N/A
10064622	12/2017	Murthy Aravalli	N/A	N/A
10076325	12/2017	Huang et al.	N/A	N/A
10085750	12/2017	Zergiebel et al.	N/A	N/A
10092290	12/2017	Yigit et al.	N/A	N/A
10092292	12/2017	Boudreaux et al.	N/A	N/A
10111657	12/2017	McCuen	N/A	N/A
10111665	12/2017	Aranyi	N/A	N/A

10117650	12/2017	Nicholas et al.	N/A	N/A
10117654	12/2017	Ingmanson et al.	N/A	N/A
10123796	12/2017	Westling et al.	N/A	N/A
10123798	12/2017	Baxter, III et al.	N/A	N/A
10135242	12/2017	Baber et al.	N/A	N/A
10143474	12/2017	Bucciaglia et al.	N/A	N/A
10172612	12/2018	Frushour	N/A	N/A
10172615	12/2018	Marczyk et al.	N/A	N/A
10172617	12/2018	Shelton, IV et al.	N/A	N/A
10175127	12/2018	Collins et al.	N/A	N/A
10180463	12/2018	Beckman et al.	N/A	N/A
10182816	12/2018	Shelton, IV et al.	N/A	N/A
10206677	12/2018	Harris et al.	N/A	N/A
10213201	12/2018	Shelton, IV et al.	N/A	N/A
10219804	12/2018	Linder et al.	N/A	N/A
10226239	12/2018	Nicholas et al.	N/A	N/A
10226250	12/2018	Beckman et al.	N/A	N/A
10226254	12/2018	Cabrera et al.	N/A	N/A
10245033	12/2018	Overmyer et al.	N/A	N/A
10245038	12/2018	Hopkins et al.	N/A	N/A
10251725	12/2018	Valentine et al.	N/A	N/A
10265065	12/2018	Shelton, IV et al.	N/A	N/A
10265066	12/2018	Measamer et al.	N/A	N/A
10265074	12/2018	Shelton, IV et al.	N/A	N/A
10271840	12/2018	Sapre	N/A	N/A
10271841	12/2018	Overmyer et al.	N/A	N/A
10285694	12/2018	Viola et al.	N/A	N/A
10285698	12/2018	Cappola et al.	N/A	N/A
10285700	12/2018	Scheib	N/A	N/A
10321907	12/2018	Shelton, IV et al.	N/A	N/A
10327776	12/2018	Harris et al.	N/A	N/A
10335150	12/2018	Shelton, IV	N/A	N/A
10349937	12/2018	Williams	N/A	A61B 17/07207
10349939	12/2018	Shelton, IV et al.	N/A	N/A
10349941	12/2018	Marczyk et al.	N/A	N/A
10368867	12/2018	Harris et al.	N/A	N/A
10390823	12/2018	Shelton, IV et al.	N/A	N/A
10390825	12/2018	Shelton, IV et al.	N/A	N/A
10390828	12/2018	Vendely et al.	N/A	N/A
10405857	12/2018	Shelton	N/A	N/A
10426468	12/2018	Contini et al.	N/A	N/A
10441279	12/2018	Shelton, IV et al.	N/A	N/A
10456129	12/2018	Shelton, IV et al.	N/A	N/A
10456130	12/2018	Cheney et al.	N/A	N/A
10463368	12/2018	Kostrzewski	N/A	N/A
10470762	12/2018	Leimbach et al.	N/A	N/A
10470911	12/2018	Thompson et al.	N/A	N/A
10478182	12/2018	Taylor	N/A	N/A
10478183	12/2018	Hess et al.	N/A	N/A

10478187	12/2018	Shelton, IV et al.	N/A	N/A
10542976	12/2019	Calderoni et al.	N/A	N/A
10548504	12/2019	Shelton, IV et al.	N/A	N/A
10548593	12/2019	Shelton, IV et al.	N/A	N/A
10548599	12/2019	Marczyk et al.	N/A	N/A
10561417	12/2019	Zergiebel et al.	N/A	N/A
10561418	12/2019	Richard et al.	N/A	N/A
10568621	12/2019	Shelton, IV et al.	N/A	N/A
10588629	12/2019	Malinouskas et al.	N/A	N/A
10595929	12/2019	Boudreaux et al.	N/A	N/A
10603128	12/2019	Zergiebel et al.	N/A	N/A
10610224	12/2019	Shelton, IV et al.	N/A	N/A
10617412	12/2019	Shelton, IV et al.	N/A	N/A
10624634	12/2019	Shelton, IV et al.	N/A	N/A
10675026	12/2019	Harris et al.	N/A	N/A
10687806	12/2019	Shelton, IV et al.	N/A	N/A
10695053	12/2019	Hess et al.	N/A	N/A
10729434	12/2019	Harris et al.	N/A	N/A
10765424	12/2019	Baxter, III et al.	N/A	N/A
10779822	12/2019	Yates et al.	N/A	N/A
10792038	12/2019	Becerra et al.	N/A	N/A
10828030	12/2019	Weir et al.	N/A	N/A
10863984	12/2019	Shelton, IV et al.	N/A	N/A
10905420	12/2020	Jasemian et al.	N/A	N/A
10905422	12/2020	Bakos et al.	N/A	N/A
10959726	12/2020	Williams et al.	N/A	N/A
11064999	12/2020	Nelson et al.	N/A	N/A
11090051	12/2020	Kostrzewski	N/A	N/A
11207065	12/2020	Harris et al.	N/A	N/A
11213294	12/2021	Shelton, IV et al.	N/A	N/A
11266409	12/2021	Huitema et al.	N/A	N/A
11278288	12/2021	Rector et al.	N/A	N/A
11439391	12/2021	Bruns et al.	N/A	N/A
11564684	12/2022	Harris et al.	N/A	N/A
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Primary Examiner: Lopez; Michelle

Background/Summary

FIELD

(1) This disclosure is generally related to surgical devices for endoscopic use and, more particularly, to surgical devices including articulation mechanisms for articulating tool assemblies.

BACKGROUND

(2) Various types of surgical devices used to endoscopically treat tissue are known in the art, and are commonly used, for example, for closure of tissue or organs in transection, resection, and anastomoses procedures, for occlusion of organs in thoracic and abdominal procedures, and for electrosurgically fusing or sealing tissue.

(3) During laparoscopic or endoscopic surgical procedures, access to a surgical site is achieved through a small incision or through a narrow cannula inserted through a small entrance wounds in a patient. Because of limited area available to access the surgical site, many endoscopic devices include mechanisms for articulating the tool assembly of the device in relation to a body portion of the device about a pivot axis.

(4) Typically, articulation mechanisms for articulating a tool assembly of a surgical stapling device include a lever that can be manipulated to articulate the tool assembly. To manipulate the lever, the force of a biasing member must be overcome. This increases the forces required to articulate the tool assembly.

(5) A continuing need exists in the art for an articulating mechanism that can be operated with minimal force.

SUMMARY

(6) This disclosure is directed to a surgical stapling device including a tool assembly, an elongate

body, and an articulation mechanism having an articulation lever and a pair of locking assemblies. The articulation lever is movable to articulate the tool assembly in relation to the elongate body about an articulation axis that is transverse to a longitudinal axis of the elongate body. The locking assemblies are configured to prevent movement of the articulation lever until one of the locking assemblies is moved from a locked position to an unlocked position. The locking assemblies are movable from their locked positions to their unlocked positions prior to movement of the articulation lever to reduce the force required to manipulate the articulation lever and to facilitate articulation in continuous fashion.

(7) Aspects of the disclosure are directed to an articulation assembly for use with a surgical device that includes a housing, an articulation lever, a first ratchet, a second ratchet, a drive member, an articulation link, a first locking assembly, and a second locking assembly. The housing defines a circular bore and a cavity and includes a base wall separating the circular bore from the cavity. The base wall defines an opening that communicates the circular bore with the cavity. The articulation lever is rotatably supported in relation to the circular bore. The first ratchet has an annular array of teeth supported within the circular bore. The second ratchet has an annular array of teeth positioned adjacent the first ratchet. The drive member includes a shaft having a first end coupled to the articulation lever and a second end. The shaft extends through the opening in the base wall of the housing. The drive member is rotatable in response to rotation of the articulation lever. The articulation link is coupled to the second end of the drive member such that rotation of the drive member causes longitudinal movement of the articulation link. The first locking assembly is supported on the articulation lever and includes a first locking member that is movable from a first position engaged with the first ratchet to a second position disengaged with the first ratchet. The first locking member is configured to prevent rotation of the articulation lever in a counterclockwise direction when the first locking member is in its first position. The second locking assembly is supported on the articulation lever and includes a second locking member that is movable from a first position engaged with the second ratchet to a second position disengaged with the second ratchet. The second locking member is configured to prevent rotation of the articulation lever in a clockwise direction when the second locking member is in its first position.

(8) In aspects of the disclosure, the first locking assembly includes a first actuator that is supported on the articulation lever, and the second locking assembly includes a second actuator that is supported on the articulation lever.

(9) In some aspects of the disclosure, the first actuator is movable to move the first locking member from its first position to its second position and the second actuator is movable to move the second locking member from its first position to its second position.

(10) In certain aspects of the disclosure, the first locking assembly includes a first slider that is positioned between the first actuator and the first locking member, and the second locking assembly includes a second slider that is positioned between the second actuator and the second locking member.

(11) In aspects of the disclosure, the first actuator is movable from a first position to a second position to move the first slider from a first position to a second position into engagement with the first locking member to move the first locking member from its first to its second position, and the second actuator is movable from a first position to a second position to move the second slider from a first position to a second position into engagement with the second locking member to move the second locking member from its first position to its second position.

(12) In some aspects of the disclosure, the first actuator and the first slider include abutting cam surfaces such that transverse movement of the first actuator causes longitudinal movement of the first slider.

(13) In certain aspects of the disclosure, the second actuator and the second slider include abutting cam surfaces such that transverse movement of the second actuator causes longitudinal movement of the second slider.

- (14) In aspects of the disclosure, the articulation assembly includes a biasing member that is positioned to urge the first and second actuators towards their first positions.
- (15) In some aspects of the disclosure, the first and second locking members are formed of a resilient material.
- (16) In certain aspects of the disclosure, the first locking member includes a first retention portion and a first ratchet finger, and the second locking member includes a second retention portion and a second ratchet finger.
- (17) In aspects of the disclosure, the first ratchet finger is engaged with the first ratchet when the first locking member is in its first position and the second ratchet finger is engaged with the second ratchet when the second locking member is in its first position.
- (18) In some aspects of the disclosure, the articulation assembly includes a first biasing member positioned to urge the first slider to its first position and a second biasing member positioned to urge the second slider to its first position.
- (19) In certain aspects of the disclosure, the second end of the drive member includes a disc-shaped member, and the articulation assembly includes a cam member and a coupling member.
- (20) In aspects of the disclosure, the cam member is supported on the disc-shaped member and coupled to the coupling member such that rotational movement of the drive member causes longitudinal movement of the coupling member.
- (21) In aspects of the disclosure, the coupling member is coupled to the articulation link such that longitudinal movement of the coupling member causes longitudinal movement of the articulation link.
- (22) In aspects of the disclosure, the tool assembly includes an anvil and a cartridge assembly.
- (23) Other aspects of the disclosure are directed to a surgical device that includes an elongate body, a tool assembly, and an articulation assembly. The elongate body defines a first longitudinal axis and has a proximal portion and a distal portion. The tool assembly defines a second longitudinal axis and is pivotably supported on the distal portion of the elongate body by a pivot member. The pivot member defines an articulation axis that is transverse to the first longitudinal axis. The tool assembly is pivotable about the articulation axis in relation to the elongate body between an articulated position in which the first and second longitudinal axes are coaxial and articulated positions in which the first and second longitudinal axes are misaligned. The articulation assembly includes a housing, an articulation lever, a first ratchet, a second ratchet, a drive member, an articulation link, a first locking assembly, and a second locking assembly. The housing defines a circular bore and a cavity and includes a base wall separating the circular bore from the cavity. The base wall defines an opening that communicates the circular bore with the cavity. The articulation lever is rotatably supported in relation to the circular bore. The first ratchet has an annular array of teeth supported within the circular bore. The second ratchet has an annular array of teeth supported adjacent the first ratchet. The drive member includes a shaft having a first end coupled to the articulation lever and a second end. The shaft extends through the opening in the base wall of the housing, and the drive member is rotatable in response to rotation of the articulation lever. The articulation link has a proximal portion and a distal portion. The proximal portion is coupled to the second end of the drive member such that rotation of the drive member causes longitudinal movement of the articulation link. The distal portion of the articulation link is coupled to the tool assembly such that longitudinal movement of the articulation link causes the tool assembly to move between the non-articulated and articulated positions. The first locking assembly is supported on the articulation lever and includes a first locking member that is movable from a first position engaged with the first ratchet to a second position disengaged with the first ratchet. The first locking member is configured to prevent rotation of the articulation lever in a counterclockwise direction when the first locking member is in its first position. The second locking assembly is supported on the articulation lever and includes a second locking member that is movable from a first position engaged with the second ratchet to a second position disengaged with the second

ratchet. The second locking member is configured to prevent rotation of the articulation lever in a clockwise direction when the second locking member is in its first position.

(24) Other aspects of the disclosure are directed to an articulation assembly for use with a surgical device that includes a housing, an articulation lever, a first ratchet, a second ratchet, a drive member, a first locking assembly, and a second locking assembly. The housing defines a circular bore. The articulation lever is rotatably supported in relation to the circular bore. The first ratchet has an annular array of teeth supported on the housing. The second ratchet has an annular array of teeth positioned adjacent the first ratchet. The drive member includes a shaft having a first end coupled to the articulation lever. The first locking assembly is supported on the articulation lever and includes a first locking member that is movable from a first position engaged with the first ratchet to a second position disengaged with the first ratchet. The first locking member is configured to prevent rotation of the articulation lever in a counterclockwise direction when the first locking member is in its first position. The second locking assembly is supported on the articulation lever and includes a second locking member that is movable from a first position engaged with the second ratchet to a second position disengaged with the second ratchet. The second locking member is configured to prevent rotation of the articulation lever in a clockwise direction when the second locking member is in its first position.

(25) Other features of the disclosure will be appreciated from the following description.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) Various aspects of this disclosure are described herein below with reference to the drawings, wherein:
- (2) FIG. 1 is a side perspective view of a surgical stapling device according to aspects of the disclosure with the surgical stapling device in a non-articulated, clamped position;
- (3) FIG. 2 is an enlarged view of the indicated area of detail shown in FIG. 1 showing an articulation assembly of the surgical stapling device with a lever of the articulation assembly removed;
- (4) FIG. 3 is an exploded perspective view of the articulation assembly shown in FIG. 2;
- (5) FIG. 3A is a cross-sectional view taken through a first body half-section of a rotation knob of the surgical stapling device shown in FIG. 1;
- (6) FIG. 3B is a perspective view from below of an articulation lever of the articulation assembly shown in FIG. 3;
- (7) FIG. 4 is a side perspective view of a first locking assembly of the articulation assembly shown in FIG. 3;
- (8) FIG. 5 is a side perspective view of a second locking assembly of the articulation assembly shown in FIG. 3;
- (9) FIG. 6 is a cross-sectional view taken through the first locking assembly of the articulation assembly shown in FIG. 2 with the articulation lever shown in phantom;
- (10) FIG. 7 is a cross-sectional view taken through the second locking assembly of the articulation assembly shown in FIG. 2 with the lever shown in phantom;
- (11) FIG. 8 is a cross-sectional view taken along section line 8-8 of FIG. 6;
- (12) FIG. 9 is a cross-sectional view taken along section line 9-9 of FIG. 6;
- (13) FIG. 10 is a cross-sectional view taken along section line 10-10 of FIG. 6;
- (14) FIG. 11 is a top view of the articulation assembly shown in FIG. 2 with the first locking assembly actuated and in an unlocked position; and
- (15) FIG. 12 is a perspective view from above of the surgical stapling device shown in FIG. 1 in an articulated position.

DETAILED DESCRIPTION

(16) The disclosed surgical device including exemplary aspects of the disclosed articulation assembly will now be described in detail with reference to the drawings in which like reference numerals designate identical or corresponding elements in each of the several views. However, it is to be understood that the aspects of the disclosure included herein are merely exemplary of the disclosure and may be embodied in various forms. Well-known functions or constructions are not described in detail to avoid obscuring the disclosure in unnecessary detail. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the disclosure in virtually any appropriately detailed structure.

(17) In this description, the term “proximal” is used generally to refer to that portion of the device that is closer to a clinician during use of the device in its customary manner, while the term “distal” is used generally to refer to that portion of the device that is farther from the clinician during use of the device in its customary manner. In addition, the term “clinician” is used generally to refer to medical personnel including doctors, nurses, surgeons, and support personnel. Further, directional terms such as front, rear, upper, lower, top, bottom, and similar terms are used to assist in understanding the description and are not intended to limit the disclosure.

(18) This disclosure is directed to a surgical device including an elongate body, a tool assembly supported on the elongate body for articulation, and an articulation assembly for selectively articulating the tool assembly in relation to the elongate body. The articulation assembly includes an articulation lever and a pair of locking assemblies. The articulation lever is movable to articulate the tool assembly in relation to the elongate body about an articulation axis. The locking assemblies prevent movement of the articulation lever until one of the locking assemblies is moved from a locked position to an unlocked position. The locking assemblies are movable from their locked positions to their unlocked positions prior to movement of the articulation lever to reduce the force required to manipulate the articulation lever and to facilitate articulation of the tool assembly in continuous fashion.

(19) FIG. 1 illustrates a surgical device shown in the form of a stapling device **10** that includes a handle assembly **12**, an elongate body or adapter assembly **14**, and a tool assembly **16**. As illustrated, the handle assembly **12** is manually actuated and includes a stationary handgrip **18**, a firing trigger **20**, a rotation knob **22**, and a retraction knob **24**. The handle assembly **12** supports the elongate body **14** and includes a body **12a** that defines the stationary handgrip **18**. The firing trigger **20** is movable in relation to the stationary handgrip **18** to actuate various functions of the tool assembly **16** via the elongate body **14** including approximation, stapling, and cutting. The handle assembly **12** supports the rotation knob **22** in a manner to facilitate rotation of the rotation knob **22** in relation to the handle assembly **12**. The rotation knob **22** supports the elongate body **14** such that rotation of the rotation knob **22** in relation to the handle assembly **12** rotates the elongate body **14** and the tool assembly **16** in relation to the handle assembly **12** about a longitudinal axis “X” defined by the elongate body **14**. The rotation knob **22** includes a body **23** that supports an articulation assembly **100** including an articulation lever **102** that is rotatable to articulate the tool assembly **16** as described in further detail below. The tool assembly **16** defines a longitudinal axis “Y-Y” (FIG. 26). The tool assembly **16** is pivotally coupled to the elongate body **14** about a pivot member **15** that defines an articulation axis “Z” to articulate between a non-articulated position in which the longitudinal axes “X” and “Y” of the elongate body **14** and tool assembly **16** are aligned with each other (FIG. 1) and articulated positions in which the longitudinal axes “X” and “Y” of the elongate body **14** and tool assembly **16** are offset from each other (FIG. 12).

(20) The tool assembly **16** is illustrated to be a stapling device and includes an anvil **16a** and a cartridge assembly **16b**. It is envisioned that the articulation assembly **100** according to aspects of this disclosure could be incorporated into a variety of different endoscopic surgical devices including but not limited to clip appliers, ligation devices, vessel sealing devices, and graspers.

(21) FIGS. 2-3B illustrate the articulation assembly **100** of the stapling device **10** (FIG. 1) which is supported on and within the body **23** of the rotation knob **22**. It is envisioned that the articulation assembly **100** need not be supported on the rotation knob but could be supported on a stationary housing of the stapling device **10**. In aspects of the disclosure, the body **23** of the rotation knob **22** includes first and second body half-sections **26** and **28** that are secured together to define a cavity **30**. The first body half-section **26** defines a circular bore **32** that receives the articulation assembly **100**. The circular bore **32** is separated from the cavity **30** by a base wall **33** (FIG. 3A) that defines an opening **33a** that extends between the circular bore **32** and the cavity **30**. An upper end of the first body half-section **26** defining the circular bore **32** includes or supports a first ratchet **34** and a second ratchet **36**. The first ratchet **34** has an annular array of ratchet teeth **34a** and the second ratchet **36** has an annular array of ratchet teeth **36a**. It is envisioned that the first and second ratchets **34** and **36** could be formed as a single ratchet member. In aspects of the disclosure, the second ratchet **36** has a circular configuration and is integrally formed with the first body half-section **26** and the first ratchet **34** has a circular configuration and is secured to the first body half-section **26** using known fastening techniques or devices, e.g., adhesives, welding, snap-connectors, or combinations thereof. It is envisioned that one or both first and second ratchets **34** and **36** can be integrally formed with the first body half-section **26** or formed separately from and secured to the first body half-section **26**. As described in further detail below, the ratchet teeth **34a** of the first ratchet **34** are angled in a direction to allow operation of the articulation assembly **100** only in a first direction, e.g., clockwise direction, and the ratchet teeth **36a** of the second ratchet **36** are angled to allow operation of the articulation assembly **100** only in a second direction, e.g., counterclockwise direction.

(22) The articulation assembly **100** includes the articulation lever **102**, first and second locking assemblies **104** and **106**, respectively, a drive member **108**, a cam member **110**, and a coupling member **112**. The articulation lever **102** includes a central hub portion **114**, a first extension **116**, and a second extension **118**. In aspects of the disclosure, the first and second extensions **116** and **118**, respectively, are axially aligned with each other and extend outwardly from the central hub portion **114**. In certain aspects of the disclosure, the first extension **116** extends distally from the central hub portion **114** and the second extension **118** extends proximally from the central hub portion **114**. The first extension **116** defines a transverse through bore **116a**. The central hub portion **114** of the articulation lever **102** defines a through bore **120** and includes an outer cylindrical wall **122**, an inner retention wall **124**, and outer retention walls **126** and **128**. The outer cylindrical wall **122** is received within the circular bore **32** of the first body half-section **26** of the rotation knob **22** and defines a cutout **130** that is aligned with a channel **132** defined in the first extension **116** of the articulation lever **102**. The channel **132** defined by the first extension **116** communicates with the transverse through bore **116a** in the articulation lever **102**.

(23) The outer retention walls **126** and **128** are positioned on opposite sides of the inner guide wall **124**. In aspects of the disclosure, the outer retention wall **126** and the inner retention wall **124** define a first stepped retention channel **134** on one side of the outer retention wall **124**, and the inner retention wall **128** and the outer retention wall **124** define a second stepped retention channel **136** on the other side of the outer retention wall **124**. The retention channels **134** and **136** are configured to help to secure the first and second locking assemblies **104** and **106** to the articulation lever **102** as described in further detail below. Other retention channel configurations are envisioned.

(24) The drive member **108** of the articulation assembly **100** includes a central shaft **140** and a disc-shaped member **142**. The central shaft **140** extends outwardly of the circular bore **32** in the rotation knob **22** and includes a first end portion **140a** that is received within the through bore **120** (FIG. 3B) in the central hub portion **114** of the articulation lever **102** and a second end portion **140b** that is fixedly secured to the disc-shaped member **142** within the cavity **30** of the rotation knob **22**. The central shaft **140** of the drive member **108** extends through the opening **33a** in the base wall **33** of

the first body half-section **26** of the rotation knob **22** such that rotation of the articulation lever **102** externally of the cavity **30** of the rotation knob **22** causes corresponding rotation of the central shaft **140** and of the disc-shaped member **142** within the cavity **30**. In aspects of the disclosure, the first end portion **140a** of the central shaft **140** of the drive member **108** is secured to the articulation lever **102** by a pin **148** that extends through openings **150** and **152** formed in the articulation lever **102** and the central shaft **140**, respectively. In certain aspects of the disclosure, the central shaft **140** of the drive member **108** defines elongate grooves **154** that receive protrusions **156** (FIG. 3B) formed within the central hub portion **114** of the articulation lever **102** to rotatably fix the articulation lever **102** to the drive member **108**.

(25) The disc-shaped member **142** of the drive member **108** can be integrally formed with the central shaft **140** or secured to the central shaft **140** using known fastening techniques or devices. The disc-shaped member **142** defines a radially extending slot **160** that extends from the central shaft **140** through an outer edge of the disc-shaped member **142**.

(26) The cam member **110** is received between the disc-shaped member **142** of the drive member **108** and the coupling member **112** and includes a first camming portion **162** (FIG. 3) and a second camming portion **164**. The first camming portion **162** is slidably received within the radially extending slot **160** of the disc-shaped member **142** of the drive member **108** and the second camming portion **164** of the cam member **110** is received in a transverse slot **166** formed in the coupling member **112**. In aspects of the disclosure, the camming member **110** includes a central flange **168** that has a diameter that is greater than the width of the transverse slot **166** of the coupling member **112** such that the cam member **110** can slide atop the coupling member **112** along the transverse slot **166** as the drive member **108** is rotated.

(27) In aspects of the disclosure, the coupling member **112** has a rectangular configuration and is supported on shoulders **170** (only one is shown) within a channel **172** (FIG. 3) defined by the second body half-section **28** of the rotation knob **22**. The channel **172** is dimensioned to restrict the coupling member **112** to longitudinal movement within the rotation knob **22**. When the articulation lever **102** is rotated to rotate the drive member **108**, the disc-shaped member **142** of the drive member **108** rotates to move the cam member **110** over a circular path of movement that includes transverse and longitudinal components. As the cam member **110** moves along the circular path of movement, the longitudinal component of movement of the cam member **110** is translated to longitudinal movement of the coupling member **112** within the channel **172** of the rotation knob **22**.

(28) The coupling member **112** includes a connector **178** that is coupled to an articulation link **180** (FIG. 3) of the stapling device **10** (FIG. 1). The articulation link **180** includes a distal portion (not shown) that is coupled to the tool assembly **16** such that longitudinal movement of the articulation link **180** causes the tool assembly (FIG. 1) to articulate about the articulation axis “Z” (FIG. 1). In aspects of the disclosure, the connector **178** includes a transversely extending finger **182** that extends through an opening **184** in a proximal portion of the articulation link **180**. It is envisioned that a variety of different connector configurations can be used to couple the coupling member **112** to the articulation link **180**. For a more detailed description of the construction and operation of a suitable articulation link, see U.S. Pat. No. 7,891,534.

(29) FIGS. 3-7 illustrate the first and second locking assemblies **104** and **106** which are supported within the channel **132** of the articulation lever **102** and extend into circular bore **32** (FIG. 2) defined by the first body half-section **26** of the rotation knob **22**. The first locking assembly **104** (FIG. 4) includes an actuator **200**, a slider **202**, and a locking member **204**. Similarly, the second locking assembly **106** (FIG. 5) includes an actuator **206**, a slider **208**, and a locking member **210**. The first and second locking assemblies **104** and **106** are substantially similar in construction and operate in substantially the same manner.

(30) The actuator **200** includes a body **212** (FIG. 4) that has a transversely extending post **214** and an angled cam surface **216**. The transversely extending post **214** extends through one side of the transverse through bore **116a** (FIG. 3) in the articulation lever **102** and can be pressed inwardly

from a non-depressed position to a depressed position into the through bore **116a** of the articulation lever **102** to move the body **212** of the actuator **200** inwardly into the channel **132** of the articulation lever **102**.

(31) The slider **202** is slidably supported within the channel **132** of the articulation lever **102** and includes an angled cam surface **220** that is in abutting relation with the angled cam surface **216** of the actuator **200**. The angled cam surfaces **216** and **220** are configured such that when the actuator **200** is pressed inwardly into the channel **132** of the articulation lever **102**, the slider **202** is moved longitudinally in a direction towards the circular bore **32** (FIG. 3) of the rotation knob **22** from a first position towards a second position. The slider **202** includes a post **222** that supports a biasing member, e.g., a coil spring **224**, that is positioned to urge the slider **202** towards its first position. In aspects of the disclosure, the coil spring **224** is positioned in compression about the post **222** between a surface **226** of the slider **202** (FIG. 4) and an inner wall **228** (FIG. 3B) of the articulation lever **102**. The slider **202** also includes an engagement surface **230**.

(32) The locking member **204** is formed of a resilient material, e.g., spring steel, and includes a retention portion **232**, an abutment portion **234**, and a ratchet finger **236**. The retention portion **232** of the locking member **204** is secured within the retention channel **134** (FIG. 3B) of the articulation lever **102**. In aspects of the disclosure, the retention portion **232** of the locking member **204** has a stepped configuration that corresponds to the configuration of the retention channel **134**. When the articulation lever **102** is secured to the central shaft **140** of the drive member **108** and the slider **202** is in its first position, the ratchet finger **236** of the locking member **204** engages a tooth **34a** of the first ratchet **34** (FIG. 3) to limit rotation of the articulation lever **102** to one direction, e.g., a clockwise direction. The ratchet finger **236** is formed of a resilient material that is configured to flex or ratchet over the teeth **34a** when the articulation lever **102** is rotated in the permitted direction. Rotation of the articulation lever **102** in the opposite direction, e.g., the counterclockwise direction, is prevented by the engagement of the ratchet finger **236** with one of the teeth **34a** of the first ratchet **34**.

(33) The abutment portion **234** of the locking member **204** is engaged with the engagement surface **230** of the slider **202**. When the slider **202** is moved from its first position to its second position by pressing the actuator **200** into the channel **132** (FIG. 3B) of the articulation lever **102**, the locking member **204** is deformed to disengage the ratchet finger **236** from the first ratchet **34** and, thus allow rotation of the articulation lever **102** in both directions.

(34) The actuator **206** includes a body **240** (FIG. 4) that has a transversely extending post **242** and an angled cam surface **244**. The transversely extending post **242** extends through the other side of the transverse through bore **116a** (FIG. 3) in the articulation lever **102** and can be pressed inwardly into the through bore **116a** from a non-depressed position to a depressed position to move the body **240** of the actuator **206** inwardly into the channel **132** of the articulation lever **102**.

(35) The slider **208** of the second locking assembly **106** is slidably supported within the channel **132** (FIG. 3B) of the articulation lever **102** and includes an angled cam surface **246** that is in abutting relation with the angled cam surface **244** of the actuator **206**. The angled cam surfaces **244** and **246** are configured such that when the actuator **206** is pressed inwardly into the channel **132** (FIG. 3B) of the articulation lever **102**, the slider **208** is moved longitudinally in a direction towards the circular bore **32** (FIG. 3) of the rotation knob **22** from a first position towards a second position. The slider **208** includes a post **248** that supports a biasing member, e.g., a coil spring **250**, that is positioned to urge the slider **208** towards its first position. In aspects of the disclosure, the coil spring **250** is positioned in compression about the post **248** between a surface **252** of the slider **208** (FIG. 5) and an inner wall **254** (FIG. 3B) of the articulation lever **102**. The slider **208** also includes an engagement surface **256**.

(36) The locking member **210** is formed of a resilient material, e.g., spring steel, and includes a retention portion **260**, an abutment portion **262**, and a ratchet finger **264**. The retention portion **260** of the locking member **210** is secured within the retention channel **136** (FIG. 3B) of the articulation

lever **102**. In aspects of the disclosure, the retention portion **260** of the locking member **210** has a stepped configuration that corresponds to the configuration of the retention channel **136**. When the articulation lever **102** is secured to the central shaft **140** of the drive member **108** and the slider **208** is in its first position, the ratchet finger **264** of the locking member **210** engages a tooth **36a** of the second ratchet **36** (FIG. 3) to limit rotation of the articulation lever **102** to one direction, e.g., a counterclockwise direction. The ratchet finger **264** is formed of a resilient material and is configured to flex or ratchet over the teeth **36a** when the articulation lever **102** is rotated in the permitted direction. Rotation of the articulation lever **102** in the opposite direction, e.g., the clockwise direction, is prevented by the engagement of the ratchet finger **264** with the second ratchet **36**.

(37) The abutment portion **262** of the locking member **210** is engaged with the engagement surface **252** of the slider **208**. When the slider **208** is moved from its first position to its second position by pressing the actuator **206** into the channel **132** (FIG. 3B) of the articulation lever **102**, the locking member **210** is deformed to disengage the ratchet finger **264** from the second ratchet **36** and, thus allow rotation of the articulation lever **102** in both directions.

(38) In aspects of the disclosure, each of the actuators **200** and **206** of the first and second locking assemblies **104** and **106**, respectively, defines a bore **200a** (FIG. 3) and **206a** (FIG. 5) that receives a biasing member, e.g., a coil spring **270**. The coil spring **270** is positioned between the actuators **200** and **206** to urge the actuators **200** and **206** to their non-depressed positions. In certain aspects of the disclosure, the posts **214** and **242** of actuators **200** and **206** include a cover or cap **276**. The caps **276** can be formed of a soft material such as rubber to facilitate more comfortable depression of the actuators **200** and **206** by a clinician.

(39) FIGS. 6-10 illustrate the articulation assembly **100** in a locked position. In the locked position, the actuators **200** and **206** of the first and second locking assemblies **104** and **106** are in non-depressed positions and the sliders **202** and **208** are in their first positions. More specifically, the coil spring **270** urges the actuators **200** and **206** of the first and second locking assemblies **104** and **106** to their non-depressed positions, and the coil springs **224** and **250** urge the sliders **202** and **208**, respectively, to their first positions. In the first positions of the sliders **202** and **208**, the ratchet finger **236** of the locking member **204** is engaged with a tooth **34a** of the first ratchet **34** to prevent rotation of the articulation lever **102** in a counterclockwise direction and the ratchet finger **264** of the locking member **210** is engaged with a tooth **36a** of the second ratchet **36** to prevent rotation of the articulation lever **102** in a clockwise direction. In this position, the articulation assembly **100** is locked out such that the articulation lever **102** cannot be rotated in either direction.

(40) FIGS. 11 and 12 illustrate the articulation assembly **100** when the first locking assembly **104** is unlocked and the articulation lever **102** is rotated in a counterclockwise direction to articulate the tool assembly **16** about the articulation axis “Z” (FIG. 1). When the actuator **200** is depressed in the direction of arrows “A” in FIG. 11 to depress the coil spring **270**, the cam surface **216** of the actuator **200** engages the cam surface **220** of the slider **202** to move the slider **202** in the direction of arrow “B” in FIG. 11. When the slider **202** moves in the direction of arrow “B”, the engagement surface **230** on the slider **202** engages the abutment portion **234** of the locking member **204** to deform the locking member **204** inwardly in the direction of arrow “C” to disengage the ratchet finger **236** from the tooth **34a** of the first ratchet **34**. Once the ratchet finger **236** is disengaged from the first ratchet **34**, the articulation lever **102** of the articulation assembly **100** can be rotated in a counterclockwise direction as indicated by arrow “D” in FIG. 12 to articulate the tool assembly **16** in the direction of arrow “E” in FIG. 12 about the articulation axis “Z” (FIG. 1). Operation of the second locking assembly **106** to articulate the tool assembly **16** in the clockwise direction is identical to operation of the first locking assembly **104** described above and will not be described in further detail herein.

(41) It is noted that when the actuator **200** is depressed to unlock the first locking assembly **104**, the second locking assembly **106** remains in a locked position to prevent movement of the articulation

lever **102** in a clockwise direction. It is envisioned that both actuators **200** and **206** could be simultaneously depressed to simultaneously unlock the first and second locking assemblies **104** and **106** to facilitate selective movement of the articulation lever **102** in either direction.

(42) Persons skilled in the art will understand that the devices and methods specifically described herein and illustrated in the accompanying drawings are non-limiting exemplary aspects of the disclosure. It is envisioned that the elements and features illustrated or described in connection with one exemplary aspect of the disclosure may be combined with the elements and features of another without departing from the scope of the disclosure. As well, one skilled in the art will appreciate further features and advantages of the disclosure based on the above-described aspects of the disclosure. Accordingly, the disclosure is not to be limited by what has been particularly shown and described, except as indicated by the appended claims.

Claims

1. An articulation assembly for use with a surgical device comprising: a housing defining a circular bore and a cavity and including a base wall separating the circular bore from the cavity, the base wall defining an opening that communicates the circular bore with the cavity; an articulation lever rotatably supported in relation to the circular bore; a first ratchet having an annular array of teeth supported within the circular bore; a second ratchet having an annular array of teeth positioned adjacent the first ratchet; a drive member including a shaft having a first end coupled to the articulation lever and a second end, the shaft extending through the opening in the base wall of the housing, and the drive member rotatable in response to rotation of the articulation lever; an articulation link coupled to the second end of the drive member such that rotation of the drive member causes longitudinal movement of the articulation link; a first locking assembly supported on the articulation lever, the first locking assembly comprising: a first locking member that is movable from a ratchet engaged position, engaged with the first ratchet, to a ratchet disengaged position disengaged with the first ratchet, with the first locking member preventing rotation of the articulation lever in a counterclockwise direction when the first locking member is in the ratchet engaged position; a first actuator supported on the articulation lever and movable from a first position to a second position; and a first slider positioned between the first actuator and the first locking member and movable from a disengaged position, disengaged from the first locking member, to an engaged position engaging the first locking member; wherein motion of the first actuator from the first position to the second position moves the first slider from the disengaged position to the engaged position; and wherein engagement of the first slider with the first locking member drives the first locking member into the ratchet engaged position to prevent counterclockwise rotation of the articulation lever; and a second locking assembly supported on the articulation lever, the second locking assembly comprising: a second locking member that is movable from a ratchet engaged position engaged with the second ratchet to a ratchet disengaged position disengaged with the second ratchet, with the second locking member preventing rotation of the articulation lever in a clockwise direction when the second locking member is in the ratchet engaged position; a second actuator supported on the articulation lever and movable from a first position to a second position; and a second slider positioned between the second actuator and the second locking member and movable from a disengaged position, disengaged from the second locking member, to an engaged position engaging the second locking member; wherein motion of the second actuator from the first position to the second position moves the second slider from the disengaged position to the engaged position; and wherein engagement of the second slider with the second locking member drives the second locking member into the ratchet engaged position to prevent clockwise rotation of the articulation lever.

2. The articulation assembly of claim 1, wherein the first actuator and the first slider include abutting cam surfaces such that transverse movement of the first actuator causes longitudinal

movement of the first slider, and wherein the second actuator and the second slider include abutting cam surfaces such that transverse movement of the second actuator causes longitudinal movement of the second slider.

3. The articulation assembly of claim 2, further including a biasing member positioned to urge the first and second actuators towards their corresponding engaged positions.

4. The articulation assembly of claim 2, wherein the first and second locking members are formed of a resilient material, the first locking member including a first retention portion and a first ratchet finger, and the second locking member including a second retention portion and a second ratchet finger, the first ratchet finger engaged with the first ratchet when the first locking member is in its ratchet engaged position and the second ratchet finger engaged with the second ratchet when the second locking member is in its ratchet engaged position.

5. The articulation assembly of claim 4, further including a first biasing member positioned to urge the first slider to its engaged position and a second biasing member positioned to urge the second slider to its engaged position.

6. The articulation assembly of claim 1, wherein the second end of the drive member includes a disc-shaped member, and the articulation assembly further includes a cam member and a coupling member, the cam member supported on the disc-shaped member and coupled to the coupling member such that rotational movement of the drive member causes longitudinal movement of the coupling member.

7. The articulation assembly of claim 6, wherein the coupling member is coupled to the articulation link such that longitudinal movement of the coupling member causes longitudinal movement of the articulation link.

8. A surgical device comprising: an elongate body defining a first longitudinal axis and having a proximal portion and a distal portion; a tool assembly defining a second longitudinal axis, the tool assembly pivotably supported on the distal portion of the elongate body by a pivot member, the pivot member defining an articulation axis that is transverse to the first longitudinal axis, the tool assembly being pivotable about the articulation axis in relation to the elongate body between a non-articulated position, in which the first and second longitudinal axes are coaxial, and an articulated position in which the first and second longitudinal axes are misaligned; and an articulation assembly including: a housing defining a circular bore and a cavity and including a base wall separating the circular bore from the cavity, the base wall defining an opening that communicates the circular bore with the cavity; an articulation lever rotatably supported in relation to the circular bore; a first ratchet having an annular array of teeth supported within the circular bore; a second ratchet having an annular array of teeth supported adjacent the first ratchet; a drive member including a shaft having a first end coupled to the articulation lever and a second end, the shaft extending through the opening in the base wall of the housing, and the drive member rotatable in response to rotation of the articulation lever; an articulation link having a proximal portion and a distal portion, the proximal portion coupled to the second end of the drive member such that rotation of the drive member causes longitudinal movement of the articulation link, the distal portion of the articulation link coupled to the tool assembly such that longitudinal movement of the articulation link causes the tool assembly to move between the non-articulated and articulated positions; a first locking assembly supported on the articulation lever, the first locking assembly comprising: a first locking member that is movable from a ratchet engaged position engaged with the first ratchet to a ratchet disengaged position disengaged with the first ratchet, with the first locking member preventing rotation of the articulation lever in a counterclockwise direction when the first locking member is in the ratchet engaged position; a first actuator supported on the articulation lever and movable from a first position to a second position; and a first slider positioned between the first actuator and the first locking member and movable from a disengaged position, disengaged from the first locking member, to an engaged position engaging the first locking member; wherein motion of the first actuator from the first position to the second position

moves the first slider from the disengaged position to the engaged position; and a second locking assembly supported on the articulation lever, the second locking assembly comprising: a second locking member that is movable from a ratchet engaged position engaged with the second ratchet to a ratchet disengaged position disengaged with the second ratchet, with the second locking member preventing rotation of the articulation lever in a clockwise direction when the second locking member is in the ratchet engaged position; a second actuator supported on the articulation lever and movable from a first position to a second position; and a second slider positioned between the second actuator and the second locking member and movable from a disengaged position, disengaged from the second locking member, to an engaged position engaging the second locking member; wherein motion of the second actuator from the first position moves the second slider from the disengaged position to the engaged position; and wherein engagement of the second slider with the second locking member drives the second locking member into the ratchet engaged position to prevent clockwise rotation of the articulation lever.

9. The surgical device of claim 8, wherein the first actuator and the first slider include abutting cam surfaces such that transverse movement of the first actuator causes longitudinal movement of the first slider, and wherein the second actuator and the second slider include abutting cam surfaces such that transverse movement of the second actuator causes longitudinal movement of the second slider.

10. The surgical device of claim 9, further including a biasing member positioned to urge the first and second actuators towards their corresponding engaged positions.

11. The surgical device of claim 9, wherein the first and second locking members are formed of a resilient material, the first locking member including a first retention portion and a first ratchet finger, and the second locking member including a second retention portion and a second ratchet finger, the first ratchet finger engaged with the first ratchet when the first locking member is in its ratchet engaged position and the second ratchet finger engaged with the second ratchet when the second locking member is in its ratchet engaged position.

12. The surgical device of claim 11, further including a first biasing member positioned to urge the first slider to its engaged position and a second biasing member positioned to urge the second slider to its engaged position.

13. The surgical device of claim 8, wherein the second end of the drive member includes a disc-shaped member, and the articulation assembly further includes a cam member and a coupling member, the cam member supported on the disc-shaped member and coupled to the coupling member such that rotational movement of the drive member causes longitudinal movement of the coupling member.

14. The surgical device of claim 13, wherein the coupling member is coupled to the proximal portion of the articulation link such that longitudinal movement of the coupling member causes longitudinal movement of the articulation link.

15. The surgical device of claim 8, wherein the tool assembly includes an anvil and a cartridge assembly.

16. An articulation assembly for use with a surgical device comprising: a housing defining a circular bore; an articulation lever rotatably supported in relation to the circular bore; a first ratchet having an annular array of teeth supported on the housing; a second ratchet having an annular array of teeth positioned adjacent the first ratchet; a drive member including a shaft having a first end coupled to the articulation lever; a first locking assembly supported on the articulation lever, the first locking assembly comprising: a first locking member that is movable from a ratchet engaged position, engaged with the first ratchet, to a ratchet disengaged position disengaged with the first ratchet, with the first locking member preventing rotation of the articulation lever in a counterclockwise direction when the first locking member is in the ratchet engaged position; a first actuator supported on the articulation lever and movable from a first position to a second position; and a first slider positioned between the first actuator and the first locking member and movable

from a disengaged position, disengaged from the first locking member, to an engaged position engaging the first locking member; wherein motion of the first actuator from the first position to the second position moves the first slider from the disengaged position to the engaged position; and wherein engagement of the first slider with the first locking member drives the first locking member into the ratchet engaged position to prevent counterclockwise rotation of the articulation lever; and a second locking assembly supported on the articulation lever, the second locking assembly comprising: a second locking member that is movable from a ratchet engaged position, engaged with the second ratchet, to a ratchet disengaged position disengaged with the second ratchet, with the second locking member preventing rotation of the articulation lever in a clockwise direction when the second locking member is in the ratchet engaged position; a second actuator supported on the articulation lever and movable from a first position to a second position; and a second slider positioned between the second actuator and the second locking member and movable from a disengaged position, disengaged from the second locking member, to an engaged position engaging the second locking member; wherein motion of the second actuator from the first position moves the second slider from the disengaged position to the engaged position; and wherein engagement of the second slider with the second locking member drives the second locking member into the ratchet engaged position to prevent clockwise rotation of the articulation lever.
